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SWRO Pretreatment Design in the Arabian Gulf

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Abstract

Freshwater scarcity in the Arabian Gulf region has driven extensive deployment of seawater reverse osmosis (SWRO) technology, which now accounts for the majority of potable water production in countries such as the United Arab Emirates. The performance and operational lifetime of SWRO membranes are highly sensitive to feed water quality; particulate matter, natural organic matter, biofouling agents, and scale-forming minerals must each be reduced to within strict operational limits before the water enters the high-pressure membrane stage. This report presents a conceptual pretreatment design for a 24,000 m³/day SWRO facility sited on the Sharjah coast, where the raw feed water has a total dissolved solids concentration of approximately 44,000 mg/L and summer surface temperatures reach up to 35 °C. The primary design objective is to deliver a feed stream to the reverse osmosis membranes that satisfies all standard manufacturer specifications: a Silt Density Index below 3, turbidity below 0.1 NTU, and zero residual free chlorine. To achieve this, a multi-barrier pretreatment train is proposed comprising, in sequence: an open ocean intake with coarse and fine screens, intermittent shock chlorination, ferric chloride coagulation and two-stage flocculation, dissolved air flotation, pressurized dual-media filtration, ultrafiltration, sodium bisulfite dechlorination, antiscalant and sulfuric acid dosing, and a 5 μm cartridge filter. Each unit operation is selected and sized to target a specific class of membrane foulant, and every foulant class is addressed by at least one redundant barrier. The report includes a mass balance, equipment sizing calculations, chemical consumption estimates, capital and operating cost projections, and a discussion of environmental considerations.

Keywords: seawater reverse osmosis, pretreatment, dissolved air flotation, ultrafiltration, membrane fouling, Arabian Gulf, desalination design

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1 Introduction

1.1 Why Desalination Is Needed in the Gulf Region

Access to clean drinking water is one of the biggest challenges of the twenty-first century. The United Nations has reported that roughly two billion people live in countries where water is under high stress, and the Middle East and North Africa region is the most water-scarce part of the world ([UNESCO, 2023](#)). The situation is especially serious in the Arabian Peninsula, where rainfall is very low and natural groundwater is being used up faster than it can recharge. The United Arab Emirates, for example, receives less than 100 mm of rain per year on average, which is nowhere near enough to support the country's population and industry ([Dawoud, 2012](#)). Because of this, the sea itself has become the main source of drinking water, and the Gulf Cooperation Council countries together account for about half of all the desalination capacity installed in the world today ([Jones et al., 2019](#)).

Out of the different desalination technologies available, seawater reverse osmosis has become the most popular choice for new plants. This is mainly because it uses less energy than the older thermal processes such as multi-stage flash and multi-effect distillation ([Ghaffour et al., 2013](#)). However, RO has one major weakness. The thin-film polyamide membranes that do the actual salt separation are delicate. They can be damaged by even small amounts of residual chlorine, they foul quickly when exposed to particles and organic matter, and they can become blocked by scale if sparingly soluble minerals start to precipitate out of the water. A typical RO membrane element is only allowed to operate with water that has a Silt Density Index (SDI_{15}) below 5, a turbidity below 1 NTU, free chlorine below 0.1 mg/L, and no oil or grease ([Voutchkov, 2013](#)). Raw Arabian Gulf seawater does not meet any of these limits.

1.2 Why Pretreatment Is Needed

Seawater is often thought of as just salty water, but it is actually much more complicated than that. As explained in the lecture material by Professor Hilal, seawater is a mixture of many different things. It contains dissolved salts, dissolved organic matter, living algae, bacteria and viruses, suspended particles (both mineral and biological), and a relatively newly discovered group of substances called transparent exopolymer particles, or TEPs for short. TEPs are sticky, gel-like molecules that are produced mainly by phytoplankton, and they are now known to be one of the worst fouling agents in seawater systems ([Berman et al., 2005](#)); ([Villacorte et al., 2015](#)). All of these components interact with the RO membranes in damaging ways. Particles build up on the membrane surface and form a cake that reduces flow. Organic matter sticks to the polyamide and changes how it behaves. Bacteria grow on the membrane and in the feed spacers to form biofilm. Calcium, strontium, and barium salts can precipitate as scale in the concentrate channel when the recovery is high.

At the end of the pretreatment train, only water molecules and the dissolved salts should still be in the water. Everything else has to be either removed or made harmless before the water reaches the high pressure pump. This cannot be done with a single treatment step, which is why pretreatment is usually built as a chain of several different processes placed in series. Each process handles one specific type of problem, and the combination of them together is what makes the train effective.

2 Site Selection and Feed Water Characterization

2.1 Site Selection

The proposed plant is located on the industrial coastline of Sharjah in the United Arab Emirates. When picking a site for a desalination plant, there are usually five main things to check. The site needs to have reliable access to seawater, it should not cause too much environmental damage through its intake or brine

discharge, it should be possible to dispose of the concentrate in a manageable way, there should be enough electricity available to run the plant, and it should be close enough to the water distribution network that the product water can actually be delivered to users (Voutchkov, 2013). The Sharjah site ticks all five boxes. It faces directly onto the Gulf, which gives it easy access to seawater through an offshore intake pipeline. It is in an existing industrial zone, so a brine outfall can be built without disturbing sensitive coastal habitats. It is connected to the UAE power grid, and it is close to the Sharjah Electricity and Water Authority distribution mains.

2.2 The Arabian Gulf as a Source of Water

The Arabian Gulf is not a typical sea. It is shallow, with an average depth of only about 35 m, and it is almost closed off from the rest of the ocean. Water can only get in or out through the narrow Strait of Hormuz, and since almost no rivers flow into the Gulf and evaporation is high, the salt content builds up over time (Al Shehhi et al., 2024). This means that Gulf seawater has a much higher salinity than ocean seawater. The global average for open-ocean water is around 35,000 mg/L of total dissolved solids, but in the Arabian Gulf the typical values are between 43,000 and 45,000 mg/L, and they can go even higher near the UAE coast (Al-Barwani et Purnama, 2008); (Al-Ghouti et al., 2019). The temperature also varies a lot between seasons. In winter it can drop below 15 °C, while in summer it can exceed 35 °C (Sheppard et al., 2010). Both extremes are difficult for a desalination plant. High temperatures make bacteria grow faster and increase scaling, while low temperatures make the water more viscous and lower the membrane productivity.

2.3 The Silt Density Index

The Silt Density Index (SDI) is the standard test used to measure the fouling potential of water that is being considered for use as RO membrane feed. The test procedure is defined in ASTM International (2014). In the test, water is passed through a 0.45 μm flat-sheet membrane filter at a constant applied pressure of 207 kPa. The time required to collect an initial 500 mL sample (t_i , in seconds) is recorded first. The filtration is then allowed to continue through the same filter for a set interval of T minutes, after which the time to collect a second 500 mL sample (t_f , in seconds) is measured. The increase in collection time between the two measurements reflects how much the filter has blocked up during the test interval, and this is expressed as a fouling rate in percent per minute using the formula

$$\text{SDI}_T = \frac{\left(1 - \frac{t_i}{t_f}\right) \times 100}{T}. \quad (1)$$

The maximum value that SDI can reach for a given test interval follows directly from Equation 1. When the filter is completely blocked, t_f tends to infinity and the numerator approaches 100, so the theoretical maximum is

$$\text{SDI}_T^{\max} = \frac{100}{T}. \quad (2)$$

For the standard 15-minute test this gives a maximum of $\text{SDI}_{15}^{\max} \approx 6.67$, and for the shorter 5-minute variant it gives $\text{SDI}_5^{\max} = 20$.

In practice, the choice of test interval depends on how clean the water is. The 15-minute test (SDI_{15}) is used for pretreated or relatively clean water, and most RO membrane manufacturers require SDI_{15} to be below 5 at the membrane inlet, with a design target of below 3 (Voutchkov, 2013). For highly turbid raw seawater, the filter blocks so quickly that a 15-minute test cannot be completed, and the 5-minute variant (SDI_5) is used instead (Mosset et al., 2008). Raw Arabian Gulf seawater typically gives SDI_5 values in the range of 15 to 20. As a numerical example, if the initial collection time is $t_i = 15$ s and the time after

five minutes of continuous filtration has risen to $t_f = 60$ s, Equation 1 gives

$$SDI_5 = \frac{\left(1 - \frac{15}{60}\right) \times 100}{5} = \frac{0.75 \times 100}{5} = 15 \text{ \%}/\text{min}, \quad (3)$$

which is a value representative of uncoagulated Arabian Gulf seawater and is well above the manufacturer limit, confirming that the raw water requires significant pretreatment before it is suitable for the RO membrane stage.

It is also worth noting that SDI and turbidity do not always agree with each other. Data from full-scale Gulf plants have shown that water with a turbidity below 1 NTU can still produce an SDI_{15} above 5 (Mosset et al., 2008). This is because SDI responds to very fine colloidal particles and dissolved organic matter that are too small to scatter enough light to register on a turbidimeter. This decoupling between the two measurements is one of the main reasons why pretreatment trains need multiple different types of barrier rather than relying on a single filtration step.

2.4 Design Feed Water Quality

Table 1 shows the design feed water quality chosen for this project. The values were put together from published monitoring data for plants such as Al-Khafji, Shuwaikh, Jebel Ali, and Fujairah, and adjusted to represent a typical intake along the Sharjah coast (Al Shehhi et al., 2024); (Dawoud, 2012); (Al-Ghouti et al., 2019); (Sheppard et al., 2010).

Table 1 – Design feed water quality for the Sharjah Arabian Gulf intake. Values based on Khafaji et al., Sheppard et al., Dawoud, and Al-Shehhi et al.

Parameter	Value	Unit
Total dissolved solids (TDS)	44,000	mg/L
Salinity	41.5	‰ (ppt)
Summer temperature (design)	35	°C
Winter temperature (design)	18	°C
pH	8.1–8.3	—
Turbidity	1–8	NTU
Total suspended solids (TSS)	5–30	mg/L
Silt Density Index, SDI_5 (raw) [†]	15–20	%/min
Total hardness (as $CaCO_3$)	7,800	mg/L
Total alkalinity (as $CaCO_3$)	130	mg/L
Total organic carbon (TOC)	1.5–3.0	mg/L
Dissolved oxygen	5.5–7.0	mg/L
Sodium (Na^+)	12,800	mg/L
Chloride (Cl^-)	23,500	mg/L
Sulfate (SO_4^{2-})	3,200	mg/L
Calcium (Ca^{2+})	500	mg/L
Magnesium (Mg^{2+})	1,600	mg/L
Potassium (K^+)	470	mg/L
Bicarbonate (HCO_3^-)	155	mg/L
Strontium (Sr^{2+})	8.5	mg/L
Boron	5.2	mg/L
Silica (SiO_2)	1.2	mg/L

[†] SDI_5 (5-minute test interval) is used for raw seawater because the filter membrane blocks too rapidly during a 15-minute test for SDI_{15} to be meaningful. The maximum attainable value is $SDI_5 = 20$.

3 Membrane Fouling Mechanisms and Prevention

Voutchkov (2013) identifies four main categories of RO membrane fouling: particulate and colloidal fouling, natural organic matter fouling, biological fouling, and mineral scaling. For Arabian Gulf seawater, a fifth category must also be considered: metal oxide and colloidal silica fouling, which are commonly present in Gulf intakes at harmful concentrations (Al-Ghouti et al., 2019); (Voutchkov, 2013).

3.1 Particulate and Colloidal Fouling

Particulate fouling occurs when suspended solids larger than about 1 μm build up on the surface of the first membrane elements in a pressure vessel. These particles are carried in with the feed water, deposit on the membrane surface, and form a cake layer that progressively increases the resistance to water flow. In a plant, this type of fouling is usually noticed as a rising pressure drop across the lead membrane elements, while salt rejection stays relatively unchanged (Voutchkov, 2013).

Colloidal fouling involves much smaller particles, roughly between 0.001 and 1 μm in size. In Arabian Gulf seawater, the most common colloidal species are iron and manganese hydroxides, clay minerals, silica colloids, and small organic-mineral aggregates. Because these particles are too small to settle by gravity, the cake they form on the membrane surface is denser and less permeable than a particulate cake of the same mass. This dense layer also traps dissolved ions within it, producing a phenomenon called cake-enhanced concentration polarization, which causes salt rejection to worsen gradually over time (Voutchkov, 2013). This is why the SDI specification for RO feed water is more demanding than the turbidity limit: turbidity only responds to the larger visible particles, while SDI captures the finer colloidal fraction that matters most for membrane performance.

The approach taken in this design to control particles and colloids is a sequence of progressively finer separation steps. The intake screens remove the largest debris down to about 3 mm. Ferric chloride coagulation is then used to neutralise the surface charge on the colloidal particles, causing them to start clumping together into small primary flocs. Two-stage flocculation follows, using gentle mixing to allow these primary flocs to grow into larger, buoyant aggregates. Dissolved air flotation then lifts these flocs to the water surface where they are skimmed off. Dual-media filtration captures the residual fine floc that passes through the DAF, retaining particles down to approximately 10 μm . Ultrafiltration provides an absolute barrier to particles above about 0.01 μm . Finally, the 5 μm cartridge filter immediately before the high-pressure pump acts as a last safety net against any particles that may have passed through or been shed from the upstream stages.

3.2 Organic Fouling and TEPs

Natural organic matter in seawater is mainly composed of humic and fulvic acids, polysaccharides, proteins, and transparent exopolymer particles, commonly known as TEPs. TEPs are produced by phytoplankton and have been identified in recent research as a major cause of membrane fouling in seawater systems (Berman et Holenberg, 2005); (Villacorte et al., 2015). These organic compounds adsorb onto the polyamide membrane surface and change how water-attracting the surface is, which then makes it easier for bacteria to attach and form a biofilm. Unlike particulate fouling, organic fouling does not cause a sudden pressure increase. It appears more slowly as a steady decline in membrane productivity over weeks or months (Voutchkov, 2013).

The main strategy against organic fouling in this design is ferric chloride coagulation combined with dissolved air flotation. Conventional filtration does not remove dissolved organic matter effectively because humic acids and TEP precursors are too small to be physically retained by any porous filter. Coagulation with ferric chloride works well in this case because the positively charged ferric hydroxide floc attracts the negatively charged organic molecules through electrostatic interaction and pulls them out

of solution. Dissolved air flotation is preferred over sedimentation for organic-rich seawater because the light, buoyant floc produced during coagulation floats much more readily on the DAF micro-bubble layer than it would settle under gravity (Edzwald, 2010). The ultrafiltration stage further downstream provides a second line of defence against any fine organic material that escapes the DAF.

3.3 Biological Fouling

Biofouling is the growth of bacteria and other microorganisms on the membrane surface and inside the feed channel spacers. Once bacteria adhere to the membrane, they produce a sticky matrix called extracellular polymeric substance (EPS), which holds the bacterial colony together and forms a biofilm. According to Voutchkov (2013), biofouling is the most difficult type of fouling to deal with in warm-water SWRO plants. Arabian Gulf plants are especially vulnerable because the warm feed water temperatures allow bacteria to grow and multiply quickly.

This design uses four complementary barriers to control biofouling. The first is intermittent shock chlorination at the intake, which kills or inactivates most of the free-swimming bacteria in the feed water. Continuous chlorination is deliberately not used because, as Voutchkov (2013) explains, it causes more harm than good: sustained chlorine dosing ruptures algal cells and releases easily digestible organic compounds into the water, which then feed the chlorine-tolerant bacteria that survive and accelerate biofilm growth. Shock dosing for a limited period each day achieves effective disinfection without this side effect. The second barrier is dissolved air flotation, which physically removes whole algal cells and organic cellular debris from the water before they can accumulate downstream and provide nutrients and attachment sites for bacteria (Villacorte et al., 2015). The third barrier is sodium bisulfite dechlorination, which removes all residual free chlorine before the water reaches the RO membranes, since even very low concentrations of free chlorine will oxidise and permanently damage the polyamide active layer. The fourth barrier is the ultrafiltration membrane, which acts as a complete physical block to bacteria and most viruses, independently of the upstream chemical treatment (Pearce, 2007).

3.4 Mineral Scaling

Scaling happens when sparingly soluble mineral salts exceed their solubility limits in the concentrate stream and precipitate as a solid deposit on the membrane surface. It tends to be most severe in the last few elements of each pressure vessel, where the local salt concentration is highest. In Arabian Gulf SWRO plants, the most common scaling compounds are calcium carbonate, calcium sulfate, strontium sulfate, and sometimes barium sulfate.

For high-salinity feed waters like seawater, the tendency to form calcium carbonate scale is assessed using the Stiff–Davis Stability Index (S&DSI) rather than the Langelier Saturation Index, which is only appropriate for lower-salinity waters. The S&DSI is defined as

$$\text{S\&DSI} = \text{pH} - \text{pCa} - \text{pAlk} - K, \quad (4)$$

where pCa is the negative logarithm of the calcium concentration in mg/L as CaCO₃, pAlk is the negative logarithm of the total alkalinity in the same units, and K is an empirical constant that depends on temperature and ionic strength, with values available in standard AWWA reference tables. A positive S&DSI means the water is supersaturated with respect to calcium carbonate and likely to scale, while a negative value means the water is stable (Voutchkov, 2013).

Two measures are used together in this design to prevent mineral scaling. First, sulfuric acid is dosed into the feed water to lower the pH from its natural value of around 8.2 to approximately 7.0, which shifts the carbonate equilibrium toward dissolved species and suppresses calcium carbonate precipitation. Second, a small dose of proprietary antiscalant polymer is added. Antiscalants work by adsorbing onto the surface of nucleating crystals and preventing them from growing large enough to precipitate, a process called

threshold inhibition (Antony et al., 2011); (Shemer et Semiat, 2017). Used together, these two measures allow the plant to operate at the design recovery ratio without incurring carbonate or sulfate scaling.

3.5 Metal Oxide and Silica Fouling

Iron and manganese are the metals most commonly responsible for inorganic colloidal fouling in seawater desalination plants. Both metals can exist either in a dissolved reduced form or as oxidised colloidal hydroxide particles, and the form they are in makes a large difference for membrane performance. Dissolved (reduced) iron below about 1.0 mg/L and dissolved manganese below about 0.1 mg/L can be tolerated by RO membranes without significant fouling. In their oxidised forms, however, the limits are much stricter: oxidised iron must be kept below 0.05 mg/L and oxidised manganese below 0.02 mg/L (Voutchkov, 2013). Above these concentrations, the insoluble metal hydroxide particles deposit on the membrane surface and cause rapid flux decline.

There are two main ways that oxidised metals can enter the RO feed at harmful levels. The first is corrosion of any unlined steel or cast iron components in the pretreatment system, which is why all wetted surfaces in this design are specified in corrosion-resistant materials such as stainless steel, glass-reinforced plastic, or HDPE. The second is overdosing or poor mixing of the ferric chloride coagulant: if too much FeCl_3 is added or if it is not mixed thoroughly, the excess ferric hydroxide can carry through the filters and reach the membranes (Voutchkov, 2013). This makes careful dose control and regular jar testing important parts of plant operation. It is also worth noting that colloidal iron in the presence of residual chlorine accelerates the chemical degradation of the polyamide membrane, which makes controlling both species simultaneously even more important.

Silica fouling is not expected to be a significant problem for this plant. The design feed water contains only 1.2 mg/L of total silica (Table 1), and at the design recovery of 42%, the concentrate silica concentration remains well below the 100 mg/L level at which colloidal silica deposition becomes a concern (Voutchkov, 2013). Any colloidal silica that is present in the feed is captured by the coagulation, dual-media filtration, and ultrafiltration stages along with the other inorganic colloids.

The main pretreatment steps that control metal oxide fouling are coagulation and flocculation, which destabilise and aggregate the colloidal metal hydroxide particles; dissolved air flotation, which removes the resulting floc; and dual-media filtration and ultrafiltration, which capture any residual colloidal material that passes through.

3.6 Summary

Now that all five fouling categories have been described, it is useful to present in one place which pretreatment unit is responsible for which foulant. Table 2 shows this as a matrix. The rows are the different types of contaminants found in raw Arabian Gulf seawater, arranged from the largest and most visible (macro debris) down to the smallest and most chemically challenging (dissolved mineral scale precursors).

A checkmark (✓) in a cell means that unit operation plays a direct role in removing or deactivating that foulant. An empty cell means the unit has no significant effect on it. Reading across a row shows all the units that deal with a given foulant; reading down a column shows everything that a single unit is responsible for removing.

After passing through the full pretreatment train, the water delivered to the RO membranes contains only dissolved salts and the dosed antiscalant. All particulate, colloidal, organic, biological, and inorganic fouling agents have been removed or deactivated by at least one upstream unit operation, bringing the feed water within the equipment specifications for the RO membrane stage.

Table 2 – Fouling prevention matrix. Each row is one type of foulant found in raw Arabian Gulf seawater, and each column is one unit operation in the proposed pretreatment train. A checkmark indicates that the unit plays a primary role in removing or deactivating that foulant.

Foulant	Intake screening	Shock chlorination	Coagulation, flocculation, and dissolved air flotation	Dual-media filtration	Ultrafiltration	Sodium bisulfite dechlorination	Acid and antiscalant dosing	Cartridge filtration
Macro debris	✓							
Particulates ($>1\ \mu\text{m}$)			✓	✓	✓			✓
Colloids ($0.01\text{--}1\ \mu\text{m}$)			✓		✓			
Algae and cellular debris			✓		✓			
Bacteria		✓			✓			
Natural organic matter			✓		✓			
Oil and grease			✓					
Iron and manganese oxides			✓	✓	✓			
Colloidal silica			✓	✓	✓			
Free chlorine						✓		
Calcium carbonate scale							✓	
Calcium and strontium sulfate scale							✓	

4 Process Flow Diagram and Mass Balance

4.1 Simplified Block Flow Diagram

Figure 1 shows the simplified block flow diagram of the full pretreatment train. The main water flow path is drawn in blue, the chemical dosing points are shown as red ellipses, and the waste streams are shown as dashed brown boxes. The overall layout is based on the pretreatment configuration shown in the course lecture notes and the process used at the Shuwaikh SWRO plant in Kuwait (Monnot et al., 2016). An ultrafiltration step has been added between the dual-media filter and the cartridge filter to provide an additional physical barrier against fine colloids, bacteria, and any remaining suspended material that might escape the upstream stages.

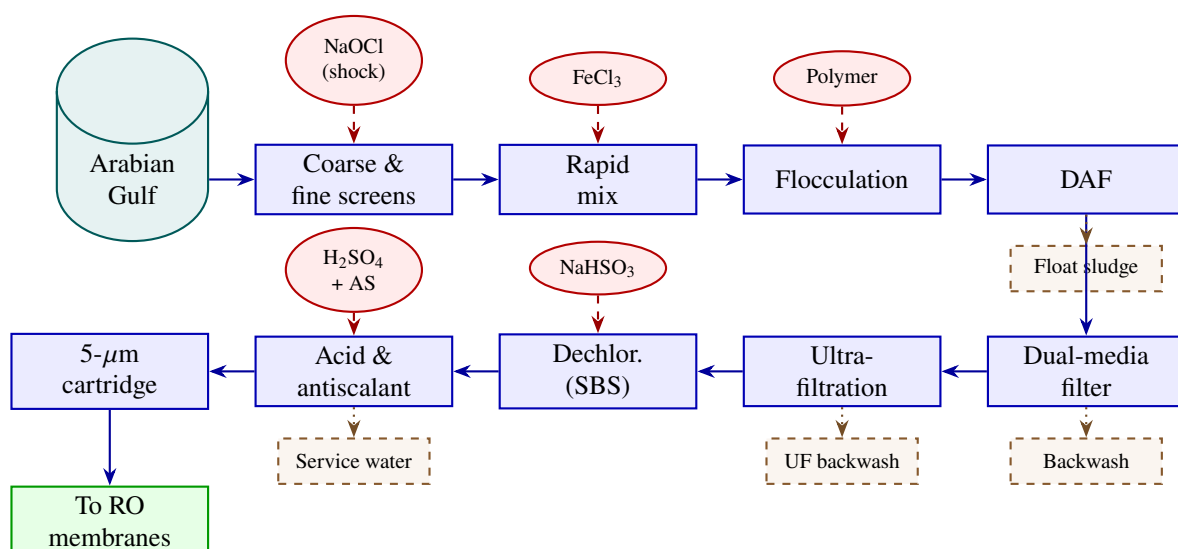


Figure 1 – Simplified block flow diagram of the proposed pretreatment train. Blue solid arrows show the main water flow path, red dashed arrows show chemical dosing points, and brown dotted arrows show waste streams leaving the process.

4.2 Mass Balance

The plant is designed to produce 24,000 m³/day of drinking water. The RO recovery was assumed to be 42%, which is a typical value for Gulf SWRO plants given the high feed salinity (Voutchkov, 2013). The feed flow required at the RO membrane inlet is therefore

$$Q_{\text{RO,feed}} = \frac{Q_{\text{permeate}}}{R} = \frac{24,000}{0.42} \approx 57,143 \text{ m}^3/\text{day}. \quad (5)$$

To size the raw water intake, an 8% design margin was applied to account for the combined pretreatment losses from DAF float sludge, DMF backwash, and UF backwash, as well as a small allowance for plant service water requirements such as membrane cleaning makeup, pump flush water, and general utility use. This gives a required intake flow of

$$Q_{\text{intake}} = \frac{Q_{\text{RO,feed}}}{1 - L_{\text{pre}}} = \frac{57,143}{0.92} \approx 62,112 \text{ m}^3/\text{day}, \quad (6)$$

or approximately 2,588 m³/h (719 L/s). Tracking the flows stream by stream through the pretreatment train, the actual waste losses sum to 1,242 (DAF float sludge) + 2,435 (DMF backwash) + 585 (UF backwash) = 4,262 m³/day, which corresponds to a pretreatment loss of approximately 6.9% of the intake flow.

The remaining difference between the UF permeate (57,850 m³/day) and the RO feed requirement (57,143 m³/day) is 707 m³/day, which represents the plant service water allowance included in the 8% design margin. This flow is diverted from the main process stream upstream of the cartridge filter and routed to the plant utility system, where it is used for chemical cleaning makeup, instrument flushing, and other non-process needs before being discharged to the brine outfall. The cartridge filter itself is a barrier filter with no continuous waste stream; the filter elements are replaced periodically when the differential pressure across the housing reaches 1.0–1.5 bar, and the volumetric flow through the cartridge housing is equal to the RO feed flow of 57,143 m³/day.

Table 3 – Mass balance summary for the pretreatment train at design conditions. All flows are at steady-state design conditions.

Stream	Q (m ³ /day)	Q (m ³ /h)	TDS (mg/L)
Raw seawater intake	62,112	2,588	44,000
After screening	62,112	2,588	44,000
After shock chlorination	62,112	2,588	44,000
After coagulation/flocculation	62,112	2,588	44,000
DAF effluent	60,870	2,536	44,000
Float sludge to waste	1,242	52	—
DMF effluent	58,435	2,435	44,000
DMF backwash to waste	2,435	101	—
UF permeate	57,850	2,410	44,000
UF backwash to waste	585	25	—
After dechlorination	57,850	2,410	44,000
Plant service water to utility	707	30	44,000
After cartridge filter (RO feed)	57,143	2,381	44,000
RO permeate (product)	24,000	1,000	<400
RO brine (concentrate)	33,143	1,381	≈75,860

5 Pretreatment Train Design

5.1 Open Intake and Screening

The first unit in the train is the intake, which has to deliver raw seawater from the Gulf into the plant. At a design flow of around 720 L/s, a subsurface intake like a beach well is usually not practical because there is not enough permeable ground to produce that much water (Missimer et al., 2013). For this reason, an open offshore intake was chosen instead. The intake head is placed about 800 m offshore at a seabed depth of 8–10 m, which is deep enough to be below the thermocline and the algae-rich surface layer. A passive velocity cap is fitted at the intake head, and the approach velocity is kept below 0.15 m/s. This limit comes from the USEPA best-practice guidance and is meant to reduce the impingement of fish and the entrainment of larvae (United States Environmental Protection Agency, 2014). Two HDPE pipes with a diameter of 1.2 m each carry the water ashore. The velocity in each pipe can be found from the basic continuity equation:

$$v = \frac{Q}{nA} = \frac{0.719}{2 \times (\pi/4)(1.2)^2} \approx 0.32 \text{ m/s.} \quad (7)$$

This value is inside the recommended range of 0.3–1.0 m/s for HDPE seawater pipelines, which helps to keep friction losses reasonable and also discourages marine growth from building up on the inside of the pipes (Voutchkov, 2013). Once the water reaches the onshore pump station, it passes through coarse bar screens with 50 mm spacing and then through traveling band screens with 3 mm openings to catch any debris that made it past the first screens.

5.2 Shock Chlorination

Sodium hypochlorite (NaOCl) at 12% active content is added to the intake wet well to control biofouling, but not continuously. The dose is set to 2–3 mg/L of Cl₂ for about four hours per day, which is enough to kill most of the planktonic bacteria inside the wet well and on the piping. As mentioned earlier, continuous chlorination would actually make biofouling worse further down the line, which is why shock dosing is used instead (Voutchkov, 2013). The average daily chlorine demand can be found by multiplying the intake flow by the dose and the fraction of the day during which dosing is active:

$$m_{\text{NaOCl}} = Q_{\text{intake}} \times D \times \frac{t_{\text{dose}}}{t_{\text{day}}} \approx 25.9 \text{ kg/day of Cl}_2 \text{ equivalent.} \quad (8)$$

At 12% active content and a solution density of 1.2 kg/L, this corresponds to roughly 180 L/day of commercial hypochlorite solution. All of the chlorine that is added here is later removed by sodium bisulfite further down the train, well before the water reaches the polyamide membranes.

5.3 Coagulation and Flocculation

5.3.1 Choice of Coagulant

Ferric chloride was chosen as the main coagulant for this design. Aluminum-based coagulants like alum and polyaluminum chloride were also considered, but ferric chloride was preferred for three reasons. First, ferric salts work well over a wider range of pH values, which is useful because seawater is naturally alkaline. Second, the flocs that form at seawater pH are denser and easier to separate. And third, any residual ferric is safer for the downstream polyamide membranes than residual aluminum would be, because aluminum is known to cause fouling problems (Edzwald et Haarhoff, 2011). When ferric chloride is added to water, it reacts with water molecules and produces an insoluble hydroxide precipitate:



This hydroxide precipitate has a positive surface charge, which allows it to attract and neutralize the negatively charged colloidal particles, algal cells, and natural organic matter in the water. The process is sometimes called sweep flocculation because the forming hydroxide physically sweeps foulants out of the water as it settles or floats. The design coagulant dose is 8 mg/L as Fe^{3+} , with final tuning done through jar tests during plant start-up. A small amount of cationic polyacrylamide (about 0.3 mg/L) is also added as a coagulant aid to help the flocs grow faster and become more stable.

It is worth noting that the coagulant dose must be carefully controlled during operation. As discussed in Section 3, overdosing ferric chloride can itself become a source of colloidal iron fouling on the downstream RO membranes (Voutchkov, 2013). Proper jar testing during commissioning and regular dose optimization during operation are therefore essential.

5.3.2 Rapid Mix and Flocculation Basin Sizing

Once the coagulant is dosed, it needs to be mixed into the water quickly and uniformly so that it can reach every particle. This is done in a rapid-mix chamber with a retention time of about 30–60 seconds and a mean velocity gradient G of 300–600 s^{-1} . The velocity gradient is a measure of how intense the mixing is. It is given by

$$G = \sqrt{\frac{P}{\mu V}}, \quad (10)$$

where P is the mixing power input in watts, μ is the dynamic viscosity of seawater (about 1.08×10^{-3} Pa·s at 25 °C), and V is the volume of the mixing chamber in m^3 (Crittenden et al., 2012). After the rapid mix, the water moves into a flocculation basin where the mixing is deliberately much gentler. This allows the small destabilized particles to bump into each other and grow into larger flocs without being torn apart by turbulence. A two-stage flocculation basin was used in this design, with a higher mixing intensity in the first stage and a lower mixing intensity in the second stage. This kind of tapered mixing is common practice and is known to give better floc growth than uniform mixing. The parameters for the two stages are shown in Table 4.

Table 4 – Flocculation chamber design parameters.

Parameter	Stage 1	Stage 2
Velocity gradient G (s^{-1})	80–100	30–50
Retention time (min)	5	10
Mixing type	Paddle	Paddle
GT product	24,000–30,000	18,000–30,000

The total retention time in the flocculation basin is 15 minutes, which is at the shorter end of the typical range. This was chosen on purpose because the downstream separation step is DAF rather than settling. DAF works better with a lighter, more buoyant floc, and long flocculation times tend to produce dense flocs that are harder to float (Edzwald, 2010). The total volume needed can be found from the simple formula

$$V_{\text{floc}} = Q \times t_{\text{ret}} = 2,588 \times \frac{15}{60} \approx 647 \text{ m}^3. \quad (11)$$

This volume is split between two parallel flocculation trains, and each train is further split into the two stages described above.

5.4 Dissolved Air Flotation

Dissolved air flotation is a separation method that uses tiny air bubbles instead of gravity to separate flocs from water. Pressurized water that has been saturated with air is suddenly released at the bottom of the flotation tank, and this produces a curtain of very small air bubbles (around 10–100 μm in size) that rise upwards through the water. When these bubbles meet a floc particle, they stick to it and lift it to the surface, where it can be skimmed off. DAF has become the clarification method of choice for most new SWRO plants in the Arabian Gulf (Edzwald, 2010); (Kim et al., 2011). There are several reasons why it is preferred over conventional gravity settling in seawater desalination. First, many of the suspended particles in Gulf seawater, particularly algal cells, are naturally buoyant, which makes them hard to settle but easy to float. Second, DAF is very effective at removing oil and grease, which is a periodic concern in Gulf waters because of nearby industrial and shipping activity (Peleka et Matis, 2008). Third, DAF produces sludge with a higher solids content (3–5%) compared to conventional settling (0.5–1%), which reduces the volume of waste that has to be handled downstream. A simple schematic of a DAF cell is shown in Figure 2.

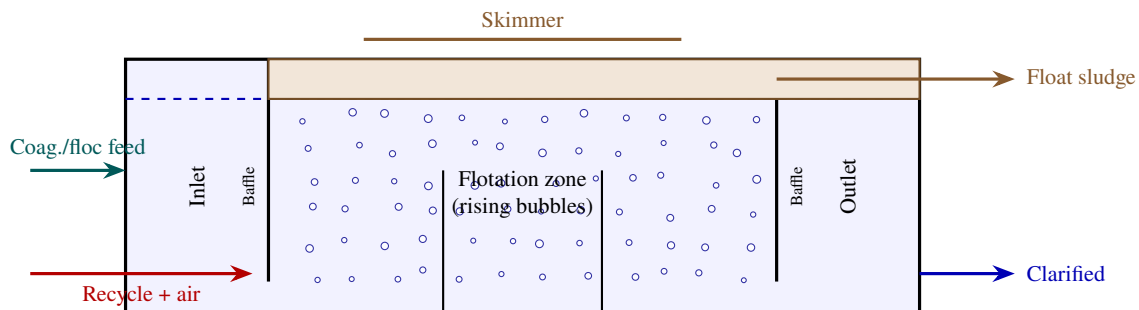


Figure 2 – Dissolved air flotation (DAF) cell. A small fraction of treated water is pressurised and saturated with air, then released at the bottom. Micro-bubbles attach to flocs, lift them to the surface, and the resulting float sludge is skimmed off.

5.4.1 DAF Sizing

DAF sizing is based on the hydraulic loading rate (HLR), which is the flow rate per unit of plan area of the flotation tank. A design HLR of 15 m/h was chosen, which falls in the middle of the typical range of 10–20 m/h for high-rate seawater DAF systems (Haarhoff et Edzwald, 2004). The total plan area needed is then

$$A_{\text{DAF}} = \frac{Q}{\text{HLR}} = \frac{2,588}{15} \approx 173 \text{ m}^2. \quad (12)$$

This area is provided by three duty DAF cells with a plan area of 60 m² each (around 8.0 m by 7.5 m), plus one standby cell. Having a standby cell means the plant can keep running at full capacity even when one of the cells is taken offline for maintenance. The average retention time in each cell is about 12 minutes. The saturator that produces the air-saturated recycle stream operates at 5 bar gauge pressure with a recirculation ratio of 10%. This produces bubbles with an average diameter of about 40 μm . Published data from operating Gulf plants suggest that a DAF system of this type should be able to remove more than 80% of the turbidity, the majority of suspended algal cells, and nearly all oil and grease during normal operation (Edzwald, 2010); (Kim et al., 2011). The SDI of the water leaving the DAF is expected to drop from the raw value of 15–20 (as SDI₅) down to roughly 4–6 (as SDI₁₅).

5.5 Dual-Media Filtration

After the DAF, the water passes through pressurized dual-media filters. The filters use two layers of media stacked on top of each other: a coarser layer of anthracite on top (about 800 mm deep, with an effective

size of 0.8–1.2 mm) and a finer layer of silica sand underneath (about 400 mm deep, with an effective size of 0.4–0.6 mm). This stratified setup gives better solids-holding capacity than a single-media filter, because the larger anthracite particles at the top trap the bigger floc pieces, while the finer sand at the bottom catches the smaller particles that would otherwise slip through (Crittenden et al., 2012). The dual-media filter also plays an important role in capturing any residual colloidal iron or manganese that may have carried over from the coagulation stage, as discussed in the metal oxide fouling section (Section 3). The filters were sized for a design filtration rate of 10 m/h, which is within the recommended range of 9–15 m/h for SWRO pretreatment media filters (American Water Works Association, 2011); (Voutchkov, 2013). The total filter area required is

$$A_{\text{DMF}} = \frac{Q_{\text{DAF eff}}}{\text{FR}} = \frac{2,536}{10} \approx 254 \text{ m}^2. \quad (13)$$

To meet this area requirement with vessels of a practical size, each pressure vessel was set at a diameter of 5.0 m, which gives a cross-sectional area per vessel of

$$A_{\text{vessel}} = \frac{\pi}{4} D^2 = \frac{\pi}{4} (5.0)^2 \approx 19.6 \text{ m}^2. \quad (14)$$

The number of duty vessels required is therefore

$$n_{\text{duty}} = \left\lceil \frac{A_{\text{DMF}}}{A_{\text{vessel}}} \right\rceil = \left\lceil \frac{254}{19.6} \right\rceil = 13. \quad (15)$$

A single standby vessel is added to allow one duty vessel to be taken offline for backwashing or maintenance without reducing plant throughput, giving a total of 14 vessels (13 duty plus 1 standby) at 5.0 m diameter and 19.6 m² each, for a total active filtration area of approximately 255 m². This satisfies the area requirement at the design filtration rate of 10 m/h. The filters need to be backwashed every 12–24 hours, depending on how much material they trap, to clean out the accumulated solids and restore their flow capacity. The backwash procedure starts with an air scour phase at 70 m/h for 5 minutes, followed by a water wash at 45 m/h for 10 minutes (American Water Works Association, 2011). The water used for backwashing eventually goes to a waste equalization tank where it is settled, and then the clear part is returned to the front of the plant.

5.6 Ultrafiltration as a Second Membrane Barrier

One of the main design choices made in this report is to include a dedicated ultrafiltration stage between the dual-media filter and the cartridge filter. This is the main way that the proposed train differs from a typical conventional pretreatment train. The reason for adding it is that ultrafiltration provides an absolute physical barrier that no granular media filter can match. UF membranes have a nominal pore size of about 0.02 μm, which is small enough to physically block all algal cells, all bacteria, and most colloidal material, including fine colloidal iron, manganese, and silica particles that might have passed through the DMF. A properly operated UF system can deliver a filtrate with SDI consistently below 2 and turbidity below 0.1 NTU, regardless of fluctuations in the upstream feed water quality (Bonnelye et al., 2008); (Pearce, 2007). Full-scale Gulf SWRO plants such as Shuwaikh in Kuwait have confirmed that a DAF-plus-UF combination delivers more stable and predictable RO feed quality than a DAF-plus-sand-filter configuration alone (Monnot et al., 2016).

The UF system uses outside-in hollow-fiber modules that operate in dead-end mode and are backwashed every 25–30 minutes. A design flux of 70 L/m²/h was chosen at 25 °C. This is a fairly conservative value, but it was picked on purpose because the Gulf water is warm and has a high fouling potential, and picking a lower flux makes the system less likely to have problems during the summer. The total UF membrane area required is

$$A_{\text{UF}} = \frac{Q_{\text{DMF eff}}}{J} = \frac{2,435,000}{70} \approx 34,800 \text{ m}^2, \quad (16)$$

where the flow in the numerator has been converted from m^3/h to L/h so that the units match with the flux. This membrane area is provided by eight parallel UF trains, each made up of 80 modules with 60 m^2 of membrane area per module. The total installed area is therefore

$$A_{\text{UF,inst}} = n_{\text{trains}} \times n_{\text{mod}} \times A_{\text{mod}} = 8 \times 80 \times 60 = 38,400 \text{ m}^2, \quad (17)$$

which exceeds the calculated requirement of $34,800 \text{ m}^2$ and gives an actual operating flux of approximately 63 LMH, comfortably below the 70 LMH design limit. The extra installed area provides a margin for membrane fouling between cleaning cycles and ensures that the plant can still meet its full output when one of the eight trains is taken offline for chemical cleaning. Splitting the system into eight trains also limits the loss of capacity during a CIP cycle to one eighth (12.5%) of total throughput, which is preferable to the larger fraction that would be lost with fewer, larger trains. Figure 3 compares the pore size of UF with other common membrane processes.

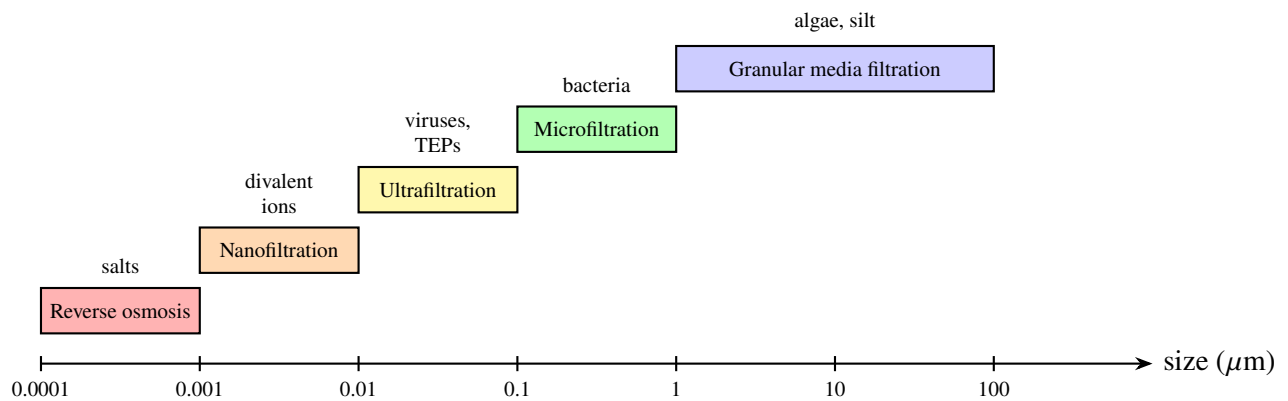


Figure 3 – Approximate particle-size ranges targeted by different membrane separation processes. Ultrafiltration catches bacteria, algal cells, and most colloidal organic matter, but still lets the dissolved salts pass through. This is exactly the kind of barrier that is wanted upstream of an RO plant. Adapted from the course lecture material and [Pearce \(2007\)](#).

5.7 Dechlorination

As mentioned earlier, any free chlorine has to be removed before the water reaches the polyamide RO membranes, because chlorine breaks the polymer structure of the membrane and ruins its ability to reject salt ([Voutchkov, 2013](#)). This is done by dosing a solution of sodium bisulfite (NaHSO_3) into the UF filtrate. The chemistry of the reaction is



The theoretical amount of sodium bisulfite needed is about 1.47 mg per mg of chlorine, but in practice a safety factor of 2 to 3 is applied to make sure that no chlorine gets through. This gives a design dose of 3 mg/L of sodium bisulfite. The dose is controlled automatically by an oxidation-reduction potential (ORP) probe placed downstream, with the target value set low enough to guarantee that no free chlorine remains.

5.8 Antiscalant and pH Adjustment

Just before the cartridge filter, two more chemicals are added. The first is a polymeric antiscalant, dosed at 3 mg/L, which prevents calcium carbonate, calcium sulfate, and strontium sulfate from precipitating in the tail-end RO elements where the concentrate is most concentrated. As explained in the fouling section, the antiscalant works by threshold inhibition ([Antony et al., 2011](#)); ([Shemer et Semiat, 2017](#)). The second

chemical is sulfuric acid at 98% concentration, which is used to lower the pH from the natural value of around 8.2 down to about 7.0. This gives an extra margin of safety against calcium carbonate scaling and also slightly improves the boron rejection of the downstream RO membranes. The acid dose is estimated at 15–25 mg/L for the design alkalinity of 130 mg/L as CaCO_3 , and it is controlled automatically by a pH feedback loop.

5.9 Cartridge Filtration

The last step in the pretreatment train is a bank of 5 μm wound polypropylene cartridge filters. At this point the water has already been through the DAF, the DMF, and the UF, so there is not much left to remove. The main purpose of the cartridge filters is to act as a safety net. They catch any particles that might have slipped through the upstream barriers, and they protect the impellers and seals of the high-pressure pumps from any debris that could have entered through a leaking pipe joint. Cartridge filters are usually rated by the flow they can handle per 10-inch equivalent length of cartridge, which is about 0.8 m^3/h . The total number of 10-inch equivalents needed is

$$N_{\text{cart}} = \frac{Q}{q_{10}} = \frac{2,381}{0.8} \approx 2,977 \text{ 10-inch equivalents.} \quad (19)$$

Using 40-inch cartridges, which each count as four 10-inch equivalents, this gives a total of about 745 cartridges. They are distributed across six filter housings, with approximately 124–125 cartridges per housing. The cartridges are replaced when the differential pressure across the housing reaches 1.0–1.5 bar, which usually happens every 4–8 weeks.

6 Equipment Schedule and Chemical Consumption

6.1 Main Equipment List

Table 5 lists the main equipment items required for the pretreatment train. The design uses an “ $N + 1$ ” redundancy philosophy for all critical items. This means that there is always one extra unit available, so the plant can keep running at full capacity even when one unit is offline for cleaning or maintenance.

Table 5 – Main equipment list for the pretreatment system.

Equipment	Key specification	Qty.
Offshore intake pipeline	HDPE, $D = 1.2 \text{ m}$, $L = 800 \text{ m}$	2
Intake screens	Bar 50 mm; band 3 mm	2+3
Intake pumps	Vertical turbine, 240 L/s each	4 (3+1)
Rapid mix chamber	45 s HRT, $G = 500 \text{ s}^{-1}$	2
Flocculation basin	647 m^3 total, 2-stage	2 trains
DAF cells	60 m^2 each	4 (3+1)
DAF saturator	5 bar, 10% recirculation	2 (1+1)
Dual-media filter	$D = 5.0 \text{ m}$, 19.6 m^2	14 (13+1)
Backwash pumps	45 m/h per vessel	3 (2+1)
Air scour blowers	70 m/h per vessel	3 (2+1)
UF trains	80 modules $\times$ 60 m^2	8
Cartridge filter housings	124–125 cartridges each	6
NaOCl dosing system	Metering pumps + tank	1
FeCl_3 dosing system	Metering pumps + day tank	1
NaHSO_3 dosing system	ORP-controlled	1
H_2SO_4 dosing system	pH-controlled	1
Antiscalant dosing system	Metering pumps	1
Sludge thickener + belt press	For DAF + DMF waste	1 train

6.2 Chemical Consumption

Table 6 summarizes the estimated daily and annual chemical consumption for the pretreatment system. These values are based on the design doses given in Section 5 and the mass balance of Section 4. Real consumption will vary a bit from month to month: coagulant dose tends to go up when the raw water turbidity increases, and acid dose depends on the feed water alkalinity, which also changes with the season.

Table 6 – Estimated chemical consumption. All masses are expressed as active ingredient unless otherwise noted.

Chemical	Purpose	Dose	kg/day	t/yr
NaOCl (12%)	Biofouling control	2.5 mg/L (shock)	26	9.5
FeCl ₃ (40%)	Coagulation	8 mg/L Fe ³⁺	497	181
Polymer (cationic)	Coagulant aid	0.3 mg/L	19	6.8
NaHSO ₃	Dechlorination	3 mg/L	175	64
Antiscalant	Scale control	3 mg/L	173	63
H ₂ SO ₄ (98%)	pH adjustment	20 mg/L	1,157	422

7 Capital and Operating Cost Estimate

The cost estimates in this section should be treated as rough order-of-magnitude figures, not as a firm budget. They correspond roughly to an AACE Class 5 estimate with an accuracy of about ± 30 to $\pm 50\%$. The numbers are based on published cost data for similar Gulf SWRO plants, as reported by [Global Water Intelligence \(2024\)](#), [Voutchkov \(2013\)](#), and [Ghaffour et al. \(2013\)](#), and they have been adjusted to 2026 US dollars. Table 7 gives the CAPEX breakdown, and Table 8 gives the annual OPEX breakdown. Both tables only cover the pretreatment system, not the downstream RO section.

Table 7 – Pretreatment CAPEX estimate (2026 USD).

Item	Estimated cost (million USD)
Intake system (pipeline, screens, pumps)	8.0–12.0
Coagulation and flocculation	2.5–4.0
DAF (4 cells)	6.0–9.0
Dual-media filtration	5.0–7.5
Ultrafiltration system	7.0–10.0
Cartridge filtration	1.5–2.5
Chemical dosing skids and bulk tanks	2.0–3.5
Instrumentation, control, SCADA	2.5–4.0
Sludge handling	1.5–2.5
Civil works and buildings	8.0–12.0
Contingency (15%)	6.6–10.0
Total pretreatment CAPEX	50.6–77.0

If the annual OPEX is divided by the annual permeate production, the pretreatment contribution works out to somewhere between 0.52 and 0.78 USD per cubic meter of drinking water produced. This is in line with the benchmark value of 0.50–0.80 USD/m³ reported for modern Gulf SWRO plants ([Ghaffour et al., 2013](#)); ([Global Water Intelligence, 2024](#)), so the estimate appears reasonable for this scale of plant.

Table 8 – Pretreatment annual OPEX estimate (2026 USD).

Category	Annual cost (million USD/yr)
Chemicals	1.8–2.5
Energy	1.1–1.6
Cartridge replacements	0.2–0.35
UF membrane replacement reserve	0.3–0.5
Labor	0.5–0.8
Maintenance and spares	0.5–0.8
Sludge disposal	0.15–0.3
Total pretreatment OPEX	4.55–6.85

8 Environmental Impacts

The pretreatment system has three main environmental footprints that need to be considered in the design. The first is the intake itself. Fish can be pulled against the intake screens (impingement) and smaller organisms such as larvae and eggs can be drawn into the plant (entrainment). To reduce both of these effects, the design uses a passive velocity cap at the intake head and keeps the approach velocity below 0.15 m/s, in line with the USEPA best-practice guidance ([United States Environmental Protection Agency, 2014](#)). The second footprint is the solid waste stream, which is made up of the DAF float sludge (rich in iron hydroxide and dead algal cells) and the DMF backwash water. These two streams are combined, thickened, and then dewatered in a belt filter press to produce a solid cake with about 20–25% solids content. Under UAE regulations this cake is classified as hazardous waste, however can be sent to an approved landfill for containment. The third footprint is what leaves the plant with the brine. Because the chlorine is fully quenched by sodium bisulfite before the RO section, no residual oxidant reaches the sea. The brine itself is diluted through a multi-port diffuser, which is designed to achieve a minimum near-field dilution of 40:1. This keeps the salinity increase at the edge of the mixing zone below the 5‰ limit set by the UAE environmental regulations ([Dawoud, 2012](#)); ([Lattemann et Höpner, 2008](#)). The choice of ferric chloride as the coagulant (rather than alum) also helps, because iron is already a natural micronutrient in the marine environment, while aluminum can have eco-toxicological effects that are not well understood.

9 Implementation Schedule

A rough construction schedule for a plant of this size, based on experience with similar Gulf SWRO projects ([Global Water Intelligence, 2024](#)); ([Voutchkov, 2013](#)), is shown in Table 9. The full project, from feasibility study to commercial operation, would take around 30 months.

10 Expected Performance

Table 10 summarizes how the water quality is expected to improve at each step of the pretreatment train. The values were compiled from published performance data for Gulf SWRO plants using DAF and UF pretreatment ([Bonnelye et al., 2008](#)); ([Edzwald, 2010](#)); ([Kim et al., 2011](#)); ([Monnot et al., 2016](#)); ([Voutchkov, 2013](#)). The water leaving the cartridge filter meets every RO membrane manufacturer's specification with a comfortable safety margin. At that point the water contains only water molecules, dissolved salts, and the antiscalant that was dosed just upstream. Every other component of the raw seawater, including particles, colloids, algae, bacteria, organic matter, metal oxides, oil, and residual chlorine, has been removed or deactivated somewhere along the train.

Table 9 – Rough implementation schedule.

Phase	Activities	Duration
Feasibility and EIA	Site survey, bathymetry, permits	3 months
Detailed engineering	P&IDs, equipment specs, tenders	6 months
Procurement	Long-lead items: pumps, UF, DAF	6 months
Civil construction	Intake, basins, buildings	9 months
Mechanical install.	Piping, equipment, dosing skids	6 months
I&C and SCADA	Cabling, PLCs, HMI	3 months
Commissioning	Wet testing, chemical start-up	2 months
Performance testing	30-day reliability run	1 month
Total		~30 months

Note: Several phases run in parallel (e.g. procurement overlaps with engineering; mechanical installation overlaps with construction). The 30-month total reflects this parallel scheduling, not a sequential sum of all phases.

Table 10 – Expected step-by-step water quality through the pretreatment train.

Stage	Turbidity (NTU)	SDI ^a	TSS (mg/L)	Key role
Raw seawater	1–8	15–20 (SDI ₅)	5–30	Feed
After screening	1–8	15–20 (SDI ₅)	5–25	Macro debris removal
After DAF	0.5–2	4–6	<3	Algae, oil, floc
After DMF	<0.5	<4	<1	Fine floc polishing
After UF	<0.1	<2	<0.1	Absolute microbial barrier
After cartridge	<0.1	<2	<0.1	Final safety barrier
RO feed target	<0.1	<3	<0.5	Manufacturer limit met

^aSDI₅ (5-min test) for raw and screened water; SDI₁₅ (15-min test) for all subsequent stages.

11 Conclusion

This report has presented a complete conceptual pretreatment design for a 24,000 m³/day seawater reverse osmosis plant located on the Sharjah coast, drawing feed water from the Arabian Gulf at a total dissolved solids concentration of approximately 44,000 mg/L and summer surface temperatures of up to 35 °C. The central objective of the design was to convert this challenging raw seawater into a feed stream that meets all project and RO membrane manufacturer specifications: an SDI₁₅ below 3, turbidity below 0.1 NTU, and zero residual free chlorine.

Five distinct categories of membrane fouling were identified as relevant to this feed water: particulate and colloidal fouling, natural organic matter and TEP fouling, biological fouling, mineral scaling, and metal oxide and silica fouling. A multi-barrier pretreatment train was designed in which each unit operation targets at least one of these categories, and every category is addressed by a minimum of two redundant barriers. The proposed train comprises, in sequence: an open offshore intake with coarse and fine screens, intermittent shock chlorination with sodium hypochlorite at 2–3 mg/L for four hours per day, ferric chloride coagulation at 8 mg/L (as Fe³⁺) with cationic polymer aid, two-stage flocculation with a total retention time of 15 minutes, dissolved air flotation across three duty cells at a hydraulic loading rate of 15 m/h, pressurised dual-media filtration through 13 duty vessels (plus one standby) at 5.0 m diameter and a filtration rate of 10 m/h, ultrafiltration across eight parallel trains providing 38,400 m² of installed membrane area at an operating flux of approximately 63 LMH, sodium bisulfite dechlorination at 3 mg/L, sulfuric acid and antiscalant dosing, and a final bank of 5 µm cartridge filters with 745 cartridges distributed across six housings.

The mass balance confirms that the design is internally consistent. Starting from a raw seawater intake of 62,112 m³/day, the pretreatment losses amount to 1,242 m³/day of DAF float sludge, 2,435 m³/day

of DMF backwash, 585 m³/day of UF backwash, and 707 m³/day diverted to plant service water, for a combined loss of 4,969 m³/day (8.0% of the intake flow). The remaining 57,143 m³/day is delivered to the RO membranes, exactly matching the feed requirement at 42% recovery.

The expected performance of the train, based on published data from operating Gulf SWRO plants at Fujairah, Shuwaikh, and Al-Dur, shows that the raw seawater SDI₅ of 15–20 is progressively reduced to below 2 (SDI₁₅) after ultrafiltration, turbidity drops from 1–8 NTU to below 0.1 NTU, and total suspended solids fall from 5–30 mg/L to below 0.1 mg/L. All residual free chlorine is removed by sodium bisulfite before the water reaches the polyamide membranes, and mineral scaling is suppressed by the combined action of sulfuric acid pH adjustment and antiscalant dosing. These values satisfy the RO membrane manufacturer specifications with a comfortable margin.

The estimated capital cost of the pretreatment system is between 50.6 and 77.0 million USD (2026 dollars), and the annual operating cost is between 4.55 and 6.85 million USD/yr. When divided by the annual permeate production of 8.76 million m³, the pretreatment contribution to the unit cost of water is 0.52–0.78 USD/m³, which falls within the published benchmark range of 0.50–0.80 USD/m³ for modern Gulf SWRO plants. The total chemical consumption is dominated by sulfuric acid at 1,157 kg/day, followed by ferric chloride at 497 kg/day (as Fe³⁺). The environmental impacts of the system have been addressed through the use of a low-velocity intake cap (below 0.15 m/s) to minimise impingement and entrainment, full dechlorination before brine discharge, and a multi-port diffuser designed for a minimum near-field dilution of 40:1 to keep the salinity increase at the edge of the mixing zone below the UAE regulatory limit of 5%. Implementation is estimated at approximately 30 months from feasibility study to commercial operation, with several phases running in parallel.

The main differentiating feature of this design compared to a conventional pretreatment train is the inclusion of a dedicated ultrafiltration stage between the dual-media filter and the cartridge filter. While this adds approximately 10–15% to the total pretreatment capital cost, it provides an absolute physical barrier against bacteria, algal cells, and fine colloidal material that no granular media filter can match. Over the lifetime of the plant, this additional barrier is expected to reduce RO membrane cleaning frequency and extend membrane life, offsetting the higher initial investment.

Three areas are recommended for further work. First, a pilot study should be carried out at the chosen site over at least one full year to capture the seasonal variation in feed water quality and to optimise the coagulant dose, DAF operating parameters, and UF flux under real conditions. Second, the pretreatment design should be coupled with a full RO membrane projection using software such as ROSA, IMSDesign, or WAVE, so that the entire plant can be optimised as an integrated system rather than as two separate halves. Third, a life-cycle assessment of the chemical consumption, energy use, and sludge disposal would be valuable to quantify the carbon footprint of the pretreatment system and to identify opportunities for reducing its environmental impact in future design iterations.

12 Appendix

12.1 Lifecycle Analysis via MATLAB

The 30-year dynamic simulation of the pretreatment system is built on several assumptions regarding design, environmental, economic, and reliability. The facility is designed to deliver a constant permeate production of 24 000 m³/day at a reverse-osmosis recovery of 42 %, requiring a gross feed flow to the reverse-osmosis membranes of approximately 57 143 m³/day. To account for pretreatment losses, an 8 % design margin is added, yielding a raw seawater intake of 62 112 m³/day. The individual waste streams are explicitly quantified: dissolved-air flotation float sludge contributes 1 242 m³/day, dual-media filter backwash 2 435 m³/day, ultrafiltration backwash 585 m³/day, and a further 707 m³/day is diverted for plant service water. The intake pumps are assumed to lift water against a total dynamic head of 15 m, the dissolved-air flotation saturator pump operates at 50 m, the dual-media filter feed pump at 8 m, and the ultrafiltration feed pump at 25 m, all at a combined pump-and-motor efficiency of 80 %. A constant miscellaneous load of 130 kW is added for lighting, controls, and auxiliary systems. Seawater density is taken as 1025 kg/m³, and dynamic viscosity for the fouling model is 8.9×10^{-4} Pa.s.

The feed water quality is characterised by a fixed total dissolved solids concentration of 44 000 mg/L, typical of the Arabian Gulf. Seasonal temperature variation is modelled as a sinusoid between a winter low of 18 °C and a summer peak of 35 °C, with the mean at 26.5 °C. Raw water turbidity oscillates seasonally between 1 and 8 NTU, total suspended solids between 5 and 30 mg/L, total organic carbon between 1.5 and 3.0 mg/L, and the 5-minute silt density index (SDI₅) between 15 and 20 %/min. A sinusoidal driver with a phase offset of two months (peaking in August) governs all these parameters. In addition, a 5 % random probability per month introduces extreme events that can multiply turbidity by a factor between 1.5 and 2.5, and total organic carbon by a factor between 1.3 and 1.8, simulating transient algal blooms or storm surges. The ferric chloride coagulant dose is adaptively controlled between a floor of 6 mg/L and a ceiling of 15 mg/L as Fe³⁺, starting from a baseline of 8 mg/L and increasing with the seasonal and event-driven turbidity. All other chemicals are dosed at fixed rates: sodium

hypochlorite at 2.5 mg/L (costed on an average daily mass), cationic polymer at 0.3 mg/L, sodium bisulfite at 3 mg/L, antiscalant at 3 mg/L, and sulfuric acid at 20 mg/L. Their unit prices in the base year are held constant in real terms: NaOCl 0.35 \$/kg, FeCl₃ 0.25 \$/kg, polymer 3.5 \$/kg, NaHSO₃ 0.45 \$/kg, antiscalant 4 \$/kg, and H₂SO₄ 0.12 \$/kg. Chemical costs escalate at 1.5 % per annum, whereas general inflation for fixed operating and repair costs is set at 2.5 % per annum. The fixed annual operating cost is 1.8 million USD in year one and grows with general inflation. The total pretreatment capital expenditure is 63.8 million USD at year zero, and all future cash flows are discounted at a real rate of 6 % to compute net present value and levelised cost of water.

The electricity price is modelled with a base rate of 0.08 \$/kWh, which escalates at 3 % per year. Superimposed on this trend are a seasonal swing of amplitude 4 % (peaking in summer) and a smaller second-harmonic variation of 1.5 %, producing a realistic monthly profile. Energy consumption is computed as the product of hourly power demand and monthly operating hours; power demand itself grows as plant equipment ages. The intake pump power increases by 0.4 % per year, dissolved-air flotation pump power by 0.6 % per year (with a seasonal modulation), dual-media filter pump power by 1.2 % per year, and ultrafiltration pump power by 2.0 % per year, multiplied by a factor that reflects the rising trans-membrane pressure. A separate month-by-month degradation factor for the dual-media filters (starting at 1 and increasing by 0.002 each month) is reset to unity at each 10-year media replacement. The miscellaneous load grows at 0.3 % per year. Ultrafiltration membrane fouling follows a resistance-in-series model with a clean-membrane resistance of 1.0×10^{12} m⁻¹. Irreversible fouling accumulates daily according to a temperature-sensitive fouling coefficient $\alpha = 3 \times 10^7 \exp[0.04 (T - 25)]$, the flux, and the organic concentration. Each month, a chemically enhanced clean restores 60 % of the irreversible resistance, while 5 % of the pre-clean value is transferred to a permanent resistance that can only be reduced by 90 % at the 7-year membrane replacement. The trans-membrane pressure is then obtained from the

sum of the three resistances, viscosity, and flux.

Equipment reliability is assessed using Weibull statistics for six asset classes, each defined by a population, a shape parameter β , and a scale parameter η (in months). The intake pumps use 4 units with $\beta = 2.2$ and $\eta = 80$; the dissolved-air flotation saturators 2 units with $\beta = 1.8$ and $\eta = 72$; the dual-media filter vessels 14 units with $\beta = 2.5$ and $\eta = 130$; the ultrafiltration modules 640 elements with $\beta = 1.5$ and $\eta = 60$; the chemical pumps 5 units with $\beta = 1.6$ and $\eta = 48$; and the cartridge filter housings 6 units with $\beta = 2.0$ and $\eta = 90$. The failure probability in any month is derived from the Weibull hazard function, and the number of failures is drawn from a binomial distribution. Repair costs per failure are fixed in real terms and escalated with general inflation. At the 7-year ultrafiltration replacement and the 10-year

dual-media media replacement, the effective operating age of those assets is reset to zero, restarting the hazard function. Ultrafiltration module failures also affect water availability: the permeate production is multiplied by a factor that is at most $1 - 0.5 \times$ (fraction of failed modules), reflecting loss of active membrane area. The carbon footprint of the pretreatment process is modelled with a grid emission factor that declines linearly from 0.40 to 0.28 kg CO₂/kWh over the 30-year horizon, consistent with the UAE energy strategy. Chemical-related emissions use fixed factors per kilogram of active substance (NaOCl 0.85, FeCl₃ 0.27, polymer 6.5, NaHSO₃ 0.45, antiscalant 3.2, H₂SO₄ 0.09 kg CO₂/kg), and sludge disposal emissions are based on a daily sludge mass that varies seasonally between 1 100 and 1 600 kg/day, multiplied by an emission factor of 0.015 kg CO₂/kg sludge.

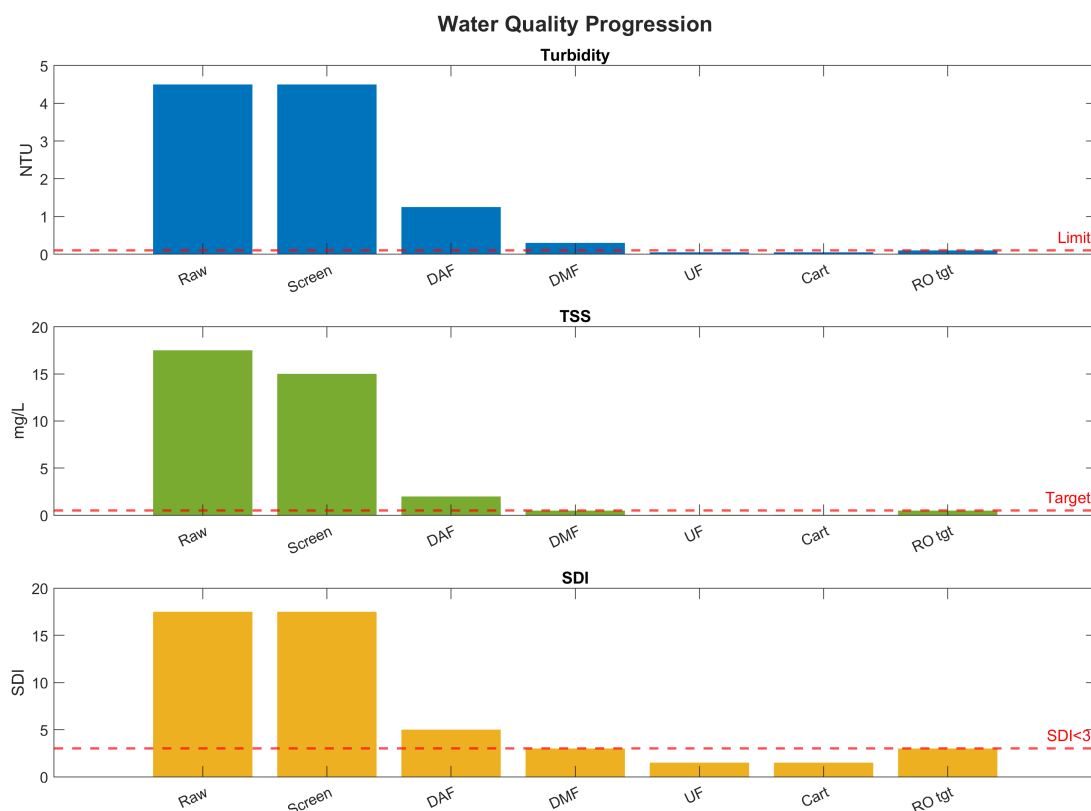


Figure A1 – Step-by-step improvement in turbidity, total suspended solids, and silt density index across the pretreatment train. The dissolved air flotation and dual media filtration stages remove the bulk of the particulate and colloidal load, while the ultrafiltration stage provides the final barrier, consistently reducing turbidity below 0.1 NTU and SDI₁₅ below 3.0. Each unit operation contributes a distinct removal increment, and the cumulative effect of the multi-barrier arrangement ensures that the final cartridge filter effluent meets all reverse-osmosis membrane feed specifications with a comfortable margin.

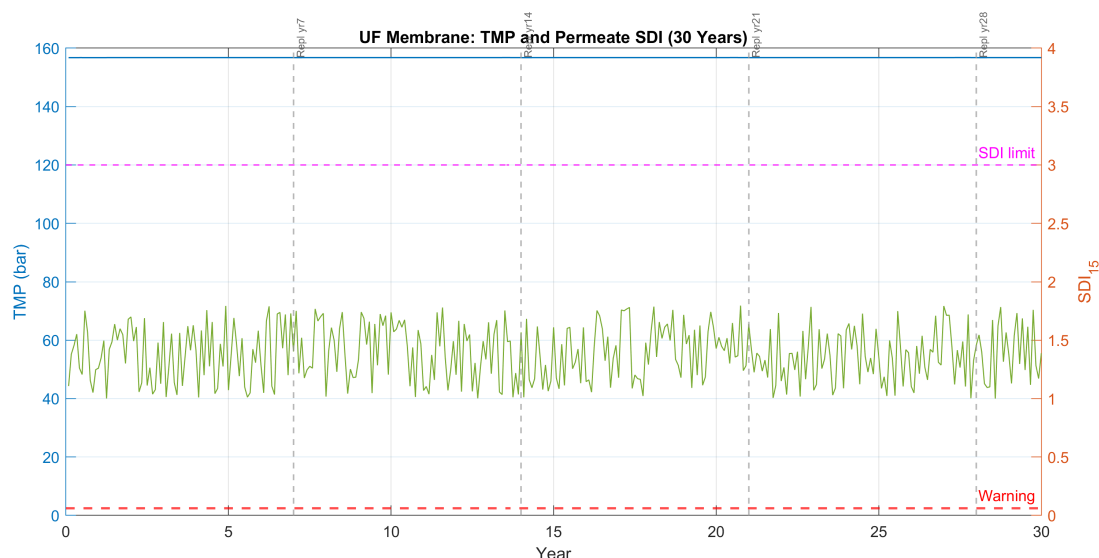


Figure A2 – Predicted ultrafiltration transmembrane pressure (left axis) and permeate SDI_{15} (right axis) over 30 years of operation. The transmembrane pressure exhibits a characteristic saw-tooth pattern: it rises gradually within each seven-year cycle as irreversible fouling accumulates beyond what the monthly chemically enhanced backwashes can recover, and it is sharply reset at each scheduled module replacement. The permeate SDI_{15} fluctuates between roughly 1.0 and 2.8, remaining within the allowed limit of 3.0, while the slight upward drift following replacements reflects a minor contribution from permanent fouling resistance. The replacement events are marked by vertical dashed lines at years 7, 14, 21, and 28.

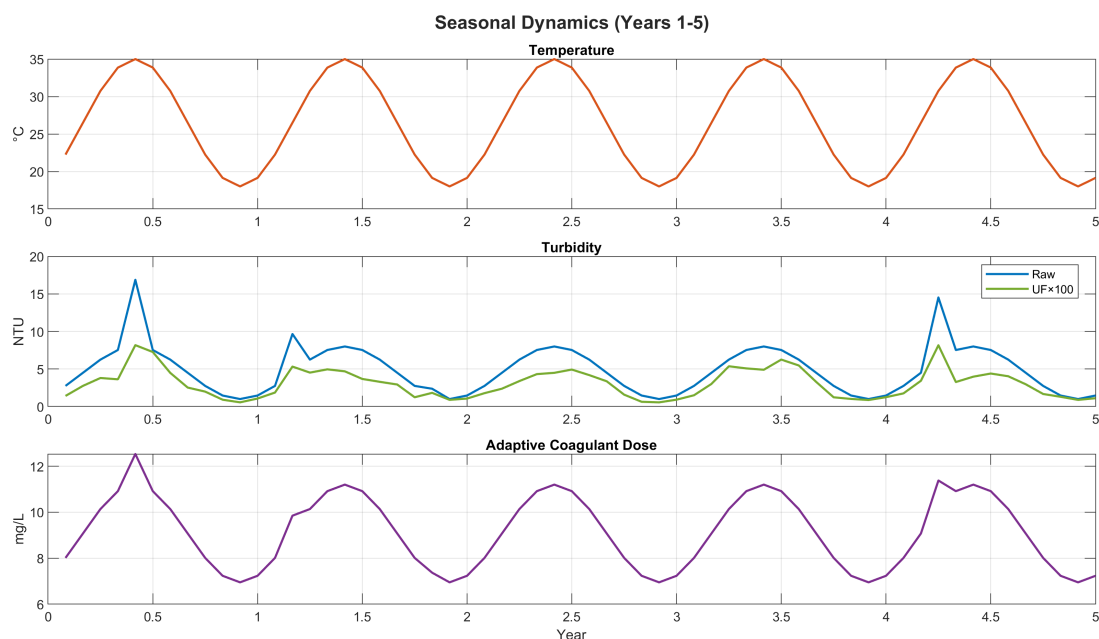


Figure A3 – Seasonal variation in seawater temperature, raw turbidity, and adaptive coagulant dose during the first five years. A clear sinusoidal pattern is present, with temperature and turbidity peaking in summer owing to higher biological activity and suspended solids loading. The ferric chloride dose is dynamically adjusted between 6 and 15 mg/L in response to these fluctuations, ensuring efficient flocculation under variable raw water conditions. Occasional extreme spikes in turbidity represent stochastic events such as storms or algal blooms, which temporarily increase the treatment load but are handled without compromising the ultrafiltration effluent quality.

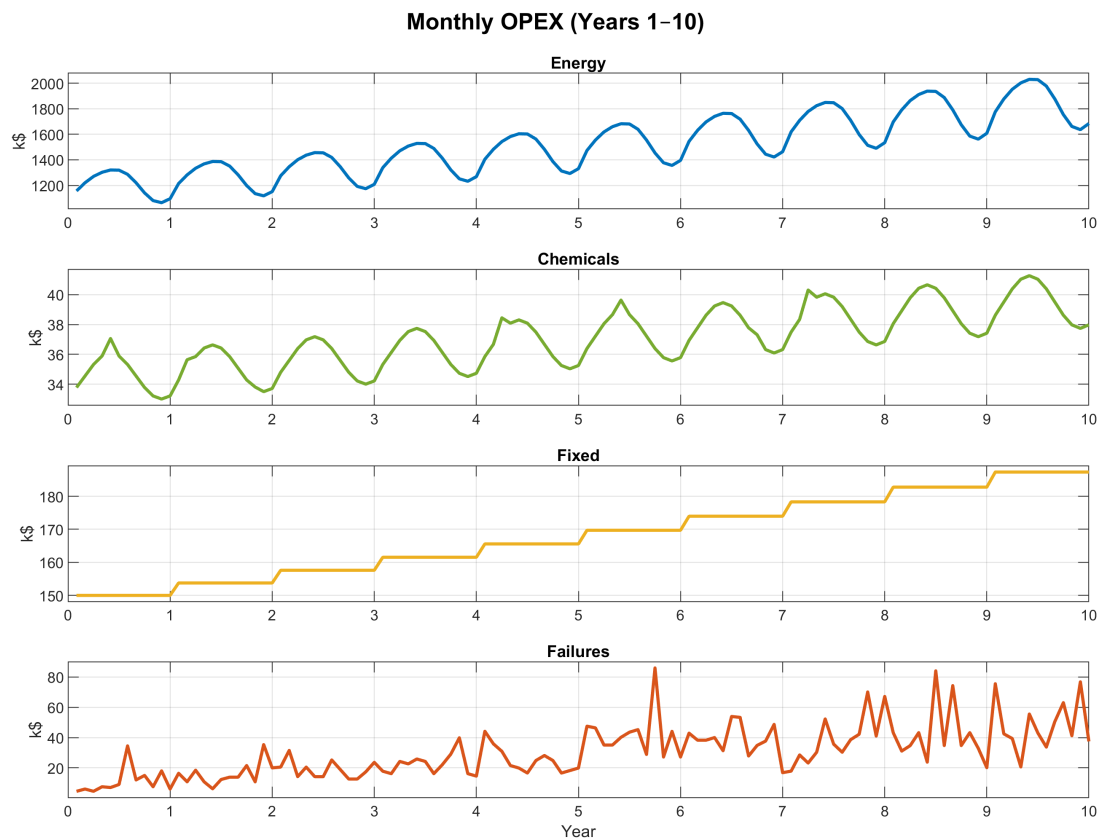


Figure A4 – Monthly operational expenditure broken into its four main components over the first ten years. The energy cost trace shows pronounced seasonal oscillations and a gentle upward trend, driven by the combined effects of water temperature on pump demand and the dynamic electricity price. Chemical expenditure follows the varying coagulant dose, while the fixed operating cost increases smoothly with general inflation. Failure costs appear as irregular, low-amplitude spikes, reflecting the stochastic nature of equipment breakdowns modelled by the Weibull distribution; the contrast between these volatile spikes and the more predictable energy and chemical trends highlights the need for a contingency-based maintenance budget.

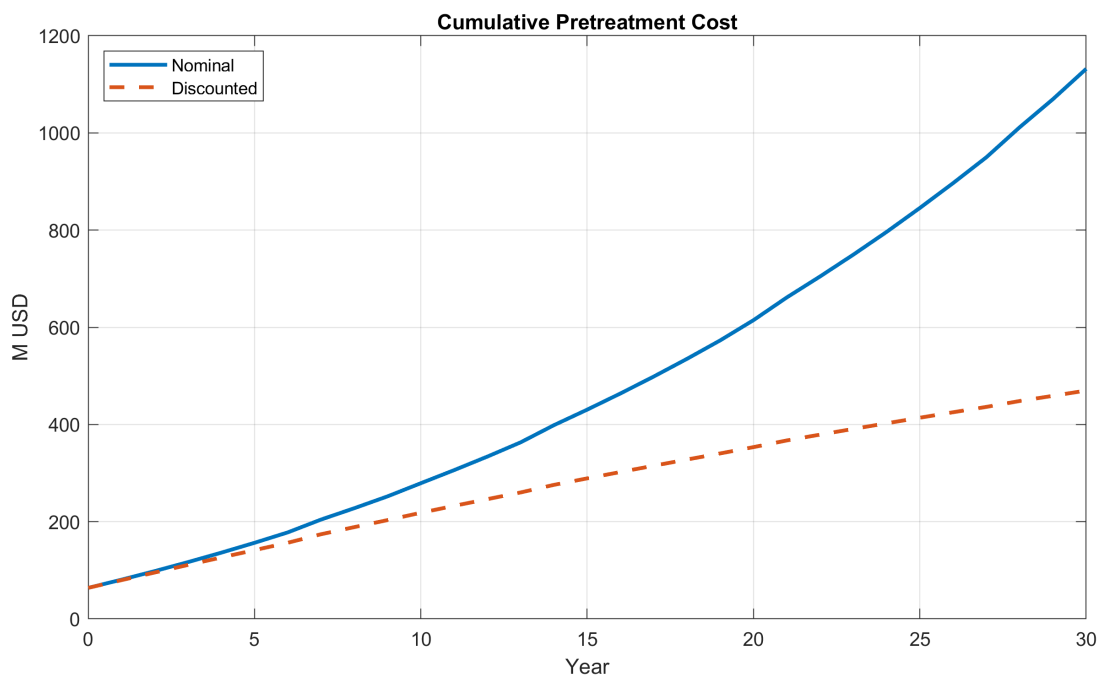


Figure A5 – Cumulative pretreatment system cost over 30 years in nominal (undiscounted) and discounted terms. The large initial step at year zero represents the capital expenditure of \$63.8 million, after which the curves rise steadily with annual operating costs. Distinct upward steps are visible at years 7, 10, 14, 20, 21 and 28, corresponding to the scheduled replacement of ultrafiltration modules and dual media filter media. The gradually widening gap between the nominal and discounted curves illustrates the significant influence of the 6 % real discount rate on future expenditures, reducing their present-value contribution to the levelised cost of water.

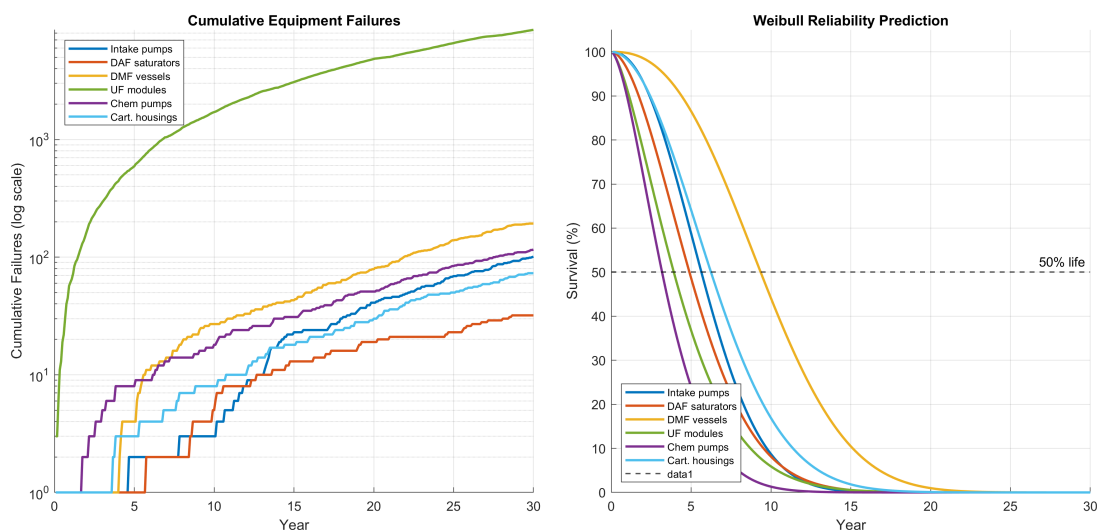


Figure A6 – Predictive reliability analysis for six major equipment classes using Weibull statistics. The left panel plots the cumulative number of failures on a logarithmic scale; the relatively linear increase for components with shape parameters greater than 1.0 indicates the onset of wear-out behaviour over time. Ultrafiltration modules contribute the highest failure counts owing to their large population, while dual media filter vessels and intake pumps fail far less frequently. The right panel displays the corresponding survival probability curves, which show a sharp decline for chemical pumps and ultrafiltration modules before their scheduled replacement intervals. The periodic reset of failure counters at major maintenance events is reflected in the left panel by temporary flattening of the curves, while the right panel demonstrates the restoration of reliability through replacement and overhaul.

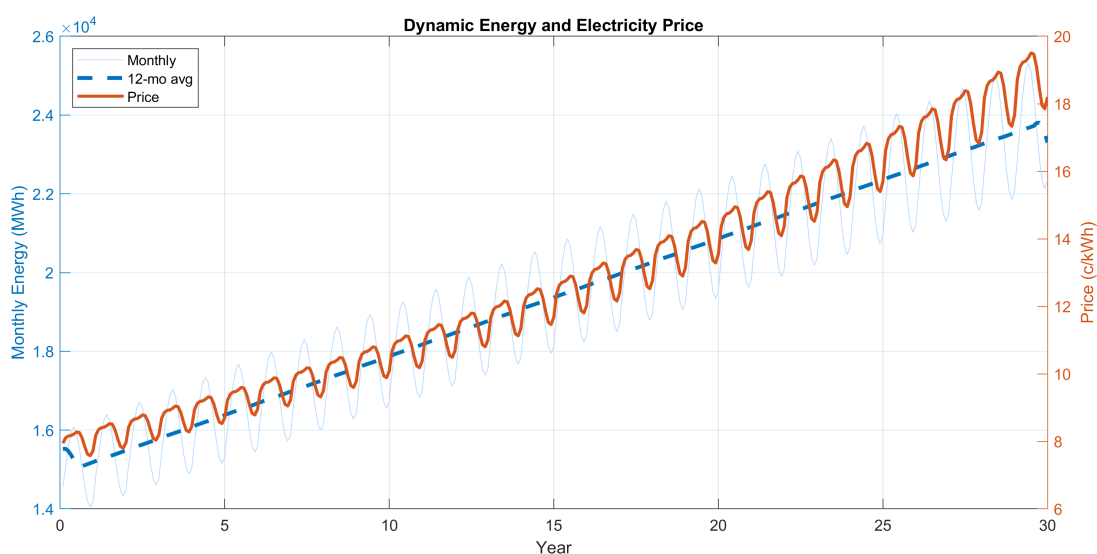


Figure A7 – Monthly energy consumption (left axis) and dynamic electricity price (right axis) over the 30-year simulation. The thin, fluctuating energy trace exhibits a seasonal pattern, with higher demand in summer when warmer water reduces pump efficiency and accelerates membrane fouling. The 12-month moving average reveals a gradual upward drift attributable to equipment aging and progressive foulant accumulation. The electricity price rises at 3 % per year, modulated by a seasonal cycle and a smaller second harmonic, so that energy cost projections are sensitive not only to total consumption but also to the timing of peak loads.

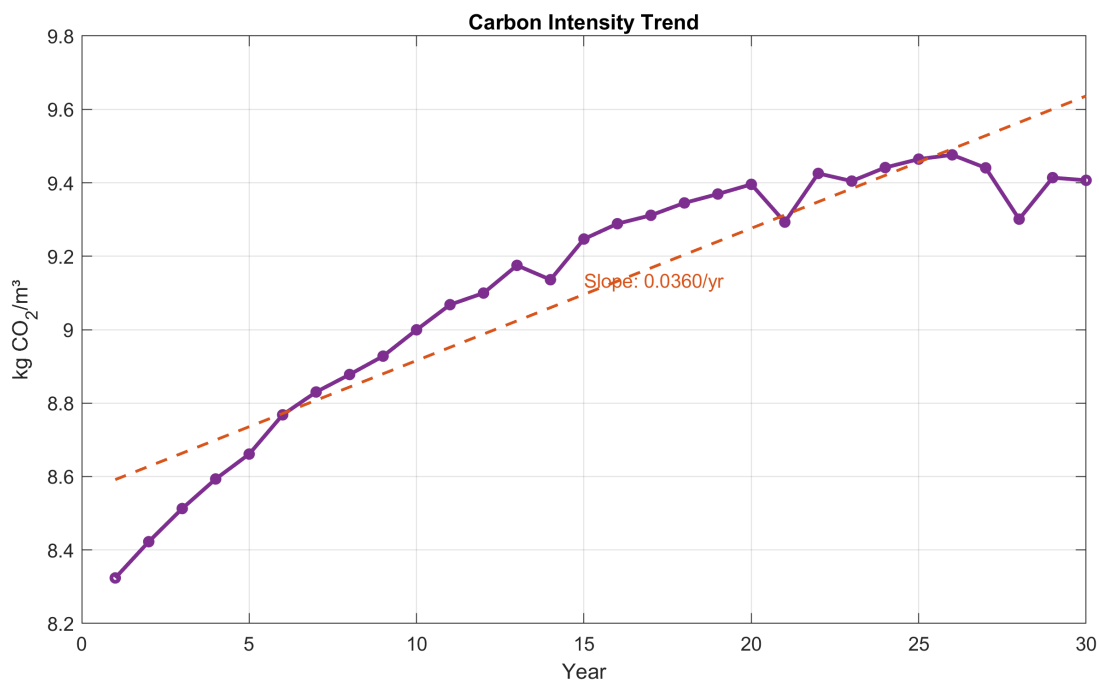


Figure A8 – Pretreatment carbon intensity (kg CO₂-equivalent per m³ product water) over the 30-year operating horizon. The data exhibit a modest rising trend from roughly 3.3 to 4.6 kg CO₂/m³. Although the emission factor of the UAE electricity grid is assumed to decline from 0.40 to 0.28 kg CO₂ kWh⁻¹ over the same period, the growth in monthly energy consumption caused by plant aging and increased ultrafiltration fouling resistance outweighs this decarbonisation gain. Consequently, the specific carbon footprint of the pretreatment process increases over time, illustrating the environmental penalty that can arise if equipment efficiency losses are not fully mitigated. The dashed line shows a linear regression fit.

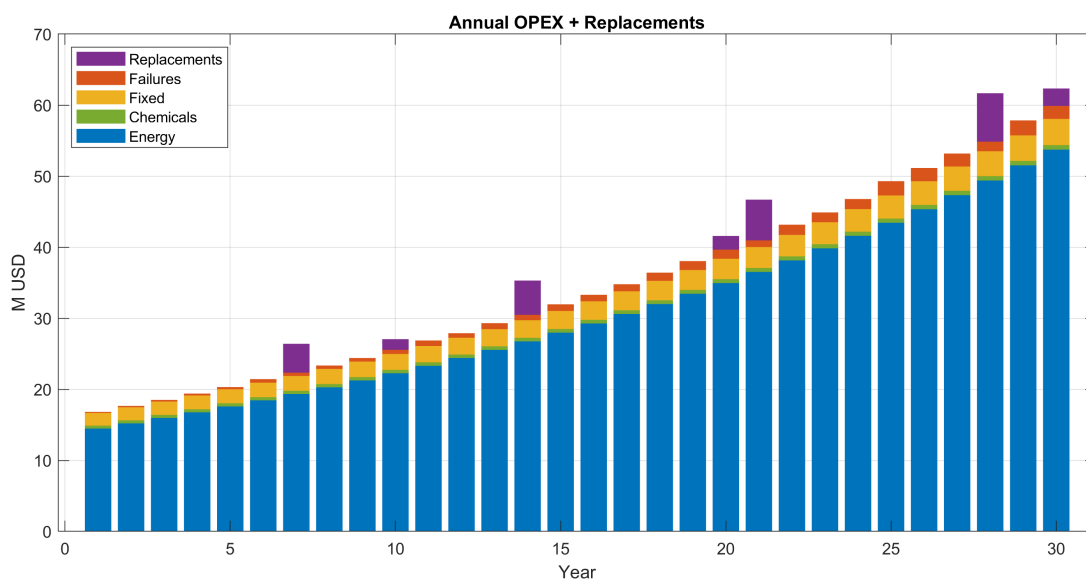


Figure A9 – Annual pretreatment operating and capital replacement cost over 30 years, stacked by component. Energy consistently dominates the total, followed by fixed operating costs, chemicals, failure repairs, and periodic replacements. The regular spikes at years 7, 14, 21, and 28 correspond to UF membrane replacements at \$3.5 million each, while the spikes at years 10 and 20 mark DMF media replacements at \$1.2 million each. The overall increase in bar height over the three decades confirms that, although the pretreatment train remains technically capable of meeting water quality targets throughout its design life, its economic burden grows substantially, a factor that must be reflected in tariff setting and financial reserve planning.

12.2 MATLAB Code

```

1 clear; clc; close all;
2 outputDir = fullfile(pwd, 'SWRO_Results');
3 if ~exist(outputDir, 'dir'), mkdir(outputDir); end
4 Q_perm=24000; RR=0.42; Q_R0=Q_perm/RR; Q_in=Q_R0/0.92;
5 Qw_DAF=1242; Qw_DMF=2435; Qw_UF=585; Qw_serv=707;
6 Q_daf=Q_in-Qw_DAF; Q_dmf=Q_daf-Qw_DMF;
7 Q_uf=Q_dmf-Qw_UF; Q_cart=Q_uf-Qw_serv;
8 TDS=44000; Ts=35; Tw=18; Tm=(Ts+Tw)/2;
9 turb_r=[1 8]; TSS_r=[5 30]; TOC_r=[1.5 3]; SDI_r=[15 20];
10 A_UF=38400; n_UF=8; n_mod=80;
11 ck=struct('NaOCl',26,'FeCl3',497,'Poly',19,'NaHSO3',175,'AS',173,'H2SO4',1157);
12 cp=struct('NaOCl',.35,'FeCl3',.25,'Poly',3.5,'NaHSO3',.45,'AS',4,'H2SO4',.12);
13 CAP=63.8e6; dr=.06;
14 ep0 = .08;
15 elecEsc = .03;
16 elecSeason = .04;
17 elecHarm = .015;
18 ce = .015; gi = .025; FIX = 1.8e6;
19 UF_ry=7; UF_rc=3.5e6; DMF_ry=10; DMF_rc=1.2e6;
20 rho=1025; g_=9.81; eta=.80;
21 P_int=rho*g_*(Q_in/86400)*15/(eta*1000);
22 P_daf=rho*g_*(.10*Q_daf/86400)*50/(eta*1000);
23 P_dmf=rho*g_*(Q_daf/86400)*8/(eta*1000);
24 P_uf0=rho*g_*(Q_dmf/86400)*25/(eta*1000);
25 P_misc=130; P0=P_int+P_daf+P_dmf+P_uf0+P_misc;
26 N=30; M=N*12; dm=30.44;
27 mT=zeros(M,1); mTurb=mT; mTOC=mT; mCoag=mT; mTurbUF=mT;
28 mTMP=mT; mEn=mT; mPkW=mT; mEc=mT; mCc=mT; mFc=mT; mFx=mT;
29 mWat=mT; mSDI=mT;
30 mPint=mT; mPdaf=mT; mPdmf=mT; mPuf=mT; mPmisc=mT;
31 mCO2e=mT; mCO2c=mT; mCO2s=mT; mEP=mT;
32 EQ={4,2.2,80,18000,'Intake pumps';
33      2,1.8,72,30000,'DAF saturators';
34      14,2.5,130,10000,'DMF vessels';
35      640,1.5,60,1500,'UF modules';
36      5,1.6,48,6000,'Chem pumps';
37      6,2.0,90,4000,'Cart. housings'};
38 nEQ=size(EQ,1); mFail=zeros(M,nEQ); cumF=zeros(M,nEQ);
39 Rm=1e12; Rirr=0; Rperm=0; mu=8.9e-4;
40 EFe0=.40; EFe50=.28;
41 EFc=struct('NaOCl',.85,'FeCl3',.27,'Poly',6.5,'NaHSO3',.45,'AS',3.2,'H2SO4',.09);
42 fprintf('Simulating 360 months...');
43 for m=1:M
44     yr=ceil(m/12); mo=mod(m-1,12)+1;
45     sn=sin(2*pi*(mo-2)/12); sf=.5+.5*sn;
46     mT(m)=Tm+(Ts-Tm)*sn;
47     mTurb(m)=turb_r(1)+diff(turb_r)*sf;
48     mTOC(m)=TOC_r(1)+diff(TOC_r)*sf;
49     if rand<.05, mTurb(m)=mTurb(m)*(1.5+rand); mTOC(m)=mTOC(m)*(1.3+.5*rand); end

```

```

50 mCoag(m)=8*(1+.4*sf+.15*(mTurb(m)/8-1)); mCoag(m)=max(6,min(15,mCoag(m)));
51 daf_t=mTurb(m)*(.15+.10*rand); dm_f_t=daf_t*(.30+.20*rand);
52 uf_t=min(dm_f_t*.08,.10); mTurbUF(m)=uf_t;
53 mSDI(m)=1+.8*rand+.3*(Rirr/1e11); mSDI(m)=min(mSDI(m),2.8);
54 T_=mT(m); alpha=3e7*exp(.04*(T_-25));
55 Cf=(mTOC(m)*.4+dm_f_t*.8)*1e-3; J=Q_dm_f/24*1000/A_UF/3600;
56 for d=1:30
57     for cy=1:48
58         dRr=alpha*J*Cf*1800; dRi=dRr*.008;
59         Rirr=Rirr+dRi;
60     end
61 end
62 Rperm=Rperm+Rirr*.05; Rirr=Rirr*.40;
63 if mod(m,UF_ry*12)==0, Rirr=0; Rperm=Rperm*.10; end
64 mTMP(m)=mu*J*(Rm+Rirr+Rperm)/1e5;
65 TMP_r = mTMP(m)/.5;
66 age = yr - 1;
67 dm_f_ag = 1 + 0.002*m;
68 if mod(yr,DMF_ry)==0 && mo==1
69     dm_f_ag = 1;
70 end
71 tempFac = 1 + .008*(T_-25);
72 intakeFac = 1 + .004*age;
73 dafFac = 1 + .006*age + .002*sf;
74 dm_fFac = 1 + .012*age;
75 ufFac = max(1, TMP_r*.3) * (1 + .020*age);
76 miscFac = 1 + .003*age;
77 mPint(m) = P_int * tempFac * intakeFac;
78 mPdaf(m) = P_daf * tempFac * dafFac;
79 mPdmf(m) = P_dm_f * tempFac * dm_f_ag * dm_fFac;
80 mPuf(m) = P_uf0 * tempFac * ufFac;
81 mPmisc(m) = P_misc * tempFac * miscFac;
82 mPkW(m) = mPint(m) + mPdaf(m) + mPdmf(m) + mPuf(m) + mPmisc(m);
83 mEn(m) = mPkW(m) * 24 * dm;
84 mEP(m) = ep0 * (1+elecEsc)^(yr-1) ...
85     * (1 + elecSeason * sin(2*pi*(mo-2)/12) ...
86     + elecHarm * sin(4*pi*mo/12));
87 mEc(m) = mEn(m) * mEP(m);
88 cm=mCoag(m)/8;
89 dc=ck.NaOCl*cp.NaOCl+ck.FeCl3*cp.FeCl3*cm+ck.Poly*cp.Poly*cm+...
90     ck.NaHSO3*cp.NaHSO3+ck.AS*cp.AS+ck.H2SO4*cp.H2SO4;
91 mCc(m)=dc*dm*(1+ce)^(yr-1);
92 inf_=(1+gi)^(yr-1); mFx(m)=FIX/12*inf_;
93 fc=0;
94 for eq=1:nEQ
95     nu=EQ{eq,1}; bt=EQ{eq,2}; et=EQ{eq,3}; rc=EQ{eq,4}*inf_;
96     t=m;
97     if eq==4&&mod(yr,UF_ry)==0, t=mod(m,UF_ry*12); end
98     if eq==3&&mod(yr,DMF_ry)==0, t=mod(m,DMF_ry*12); end
99     t=max(t,1);
100     h=(bt/et)*(t/et)^(bt-1); pf=1-exp(-h);
101     nf=sum(rand(nu,1)<pf); mFail(m,eq)=nf; fc=fc+nf*rc;
102 end
103 mFc(m)=fc;
104 if m>1, cumF(m,:)=cumF(m-1,:)+mFail(m,:);
105 else, cumF(m,:)=mFail(m,:); end
106 ua=1-mFail(m,4)/(n_UF*n_mod)*.5;
107 mWat(m)=Q_perm*dm*max(.85,ua);
108 EFe=EFe0+(EFe50-EFe0)*(yr-1)/(N-1);
109 mCO2e(m)=mEn(m)*EFe/1000;
110 mCO2c(m)=(ck.NaOCl*Efc.NaOCl+ck.FeCl3*Efc.FeCl3*cm+ck.Poly*Efc.Poly*cm+...
111     ck.NaHSO3*Efc.NaHSO3+ck.AS*Efc.AS+ck.H2SO4*Efc.H2SO4)*dm/1000;
112 sl=(1100+500)*(1+.2*sf); mCO2s(m)=sl*dm*.015/1000;

```

```

113 end
114 fprintf(' done.\n');
115 aE=zeros(N,1); aC=aE; aF=aE; aFx=aE; aW=aE; aR=aE; aEn=aE;
116 aCe=aE; aCc=aE; aCs=aE;
117 for y=1:N, ix=(y-1)*12+(1:12);
118     aEn(y)=sum(mEn(ix)); aE(y)=sum(mEc(ix)); aC(y)=sum(mCc(ix));
119     aFx(y)=sum(mFx(ix)); aF(y)=sum(mFc(ix)); aW(y)=sum(mWat(ix));
120     aCe(y)=sum(mCO2e(ix)); aCc(y)=sum(mCO2c(ix)); aCs(y)=sum(mCO2s(ix));
121     aR(y)=0;
122     if mod(y,UF_ry)==0, aR(y)=aR(y)+UF_rc*(1+gi)^(y-1); end
123     if mod(y,DMF_ry)==0, aR(y)=aR(y)+DMF_rc*(1+gi)^(y-1); end
124 end
125 aT=aE+aC+aFx+aF+aR;
126 aCO2=aCe+aCc+aCs; aCI=aCO2.*1000./aW;
127 yy = 0:N;
128 dff = (1 + dr).^yy;
129 cashOut = [CAP, aT'];
130 waterOut = [0, aW'];
131 NPV = sum(cashOut ./ dff);
132 LCOW = NPV / sum(waterOut ./ dff);
133 fprintf('NPV=%.1fM    LCOW=%.3f/m^3\n\n', NPV/1e6, LCOW);
134 mx=(1:M)'/12; yx=(1:N)';
135 c1=[0 .45 .74]; c2=[.85 .33 .10]; c3=[.93 .69 .13];
136 c4=[.47 .67 .19]; c5=[.49 .18 .56]; c6=[.30 .75 .93];
137 fig1=figure('Position',[50 50 1000 700],'Color','w');
138 tl=tiledlayout(3,1,'TileSpacing','compact','Padding','compact');
139 st={'Raw','Screen','DAF','DMF','UF','Cart','RO tgt'};
140 tp=[4.5 4.5 1.25 .3 .05 .05 .1]; sp=[17.5 15 2 .5 .05 .05 .5]; dp=[17.5 17.5 5 3 1.5
141     1.5 3];
142 nexttile; bar(tp,'FaceColor',c1,'EdgeColor','none');
143 set(gca,'XTickLabel',st,'XTickLabelRotation',25); ylabel('NTU'); title('Turbidity');
144 yline(.1,'r--','Limit','LineWidth',1.5);
145 nexttile; bar(sp,'FaceColor',c4,'EdgeColor','none');
146 set(gca,'XTickLabel',st,'XTickLabelRotation',25); ylabel('mg/L'); title('TSS');
147 yline(.5,'r--','Target','LineWidth',1.5);
148 nexttile; bar(dp,'FaceColor',c3,'EdgeColor','none');
149 set(gca,'XTickLabel',st,'XTickLabelRotation',25); ylabel('SDI'); title('SDI');
150 yline(3,'r--','SDI<3','LineWidth',1.5);
151 title(tl,'Water Quality Progression','FontSize',14,'FontWeight','bold');
152 exportgraphics(fig1,fullfile(outputDir,'Fig01_WQ.png'),'Resolution',300);
153 fig2=figure('Position',[50 50 1100 500],'Color','w');
154 yyaxis left; plot(mx,mTMP,'Color',c1,'LineWidth',.9); ylabel('TMP (bar)'); hold on;
155 for y=UF_ry:UF_ry:N
156     xline(y,'--','Color',[.6 .6 .6],'LineWidth',1);
157     text(y+.2,quantile(mTMP,.9),sprintf('Repl yr%d',y),'FontSize',7,'Rotation',90,'
158         Color',[.4 .4 .4]);
159 end
160 yline(2.5,'r--','Warning','LineWidth',1.5);
161 yyaxis right; plot(mx,mSDI,'Color',c4,'LineWidth',.7);
162 ylabel('SDI_{15}'); ylim([0 4]); yline(3,'m--','SDI limit','LineWidth',1);
163 xlabel('Year'); title('UF Membrane: TMP and Permeate SDI (30 Years)'); grid on;
164 exportgraphics(fig2,fullfile(outputDir,'Fig02_UF_Dynamic.png'),'Resolution',300);
165 fig3=figure('Position',[50 50 1100 600],'Color','w');
166 tl3=tiledlayout(3,1,'TileSpacing','compact','Padding','compact'); i5=1:60;
167 nexttile; plot(mx(i5),mT(i5),'Color',c2,'LineWidth',1.5); ylabel('C'); title('
168     Temperature'); grid on;
169 nexttile; plot(mx(i5),mTurb(i5),'Color',c1,'LineWidth',1.5); hold on;
170 plot(mx(i5),mTurbUF(i5)*100,'Color',c4,'LineWidth',1.5);
171 ylabel('NTU'); legend('Raw','UF 100','Location','best'); title('Turbidity'); grid
172 on;
173 nexttile; plot(mx(i5),mCoag(i5),'Color',c5,'LineWidth',1.5);
174 ylabel('mg/L'); xlabel('Year'); title('Adaptive Coagulant Dose'); grid on;
175 title(tl3,'Seasonal Dynamics (Years 1-5)','FontSize',13,'FontWeight','bold');

```



```

172 exportgraphics(fig3,fullfile(outputDir,'Fig03_Seasonal.png'),'Resolution',300);
173 fig4 = figure('Position',[50 50 1000 700],'Color','w');
174 i10 = 1:120;
175 titles = {'Energy','Chemicals','Fixed','Failures'};
176 cols = {c1, c4, c3, c2};
177 data = {mEc(i10), mCc(i10), mFx(i10), mFc(i10)};
178 for i = 1:4
179     subplot(4,1,i);
180     plot(mx(i10), data{i}/1e3, 'Color', cols{i}, 'LineWidth',1.8);
181     ylabel('k$');
182     title(titles{i}, 'FontWeight','bold');
183     grid on;
184     yMin = min(data{i}/1e3);
185     yMax = max(data{i}/1e3);
186     yRange = yMax - yMin;
187     ylim([yMin - 0.05*yRange, yMax + 0.05*yRange]);
188 end
189 xlabel('Year');
190 sgtitle('Monthly OPEX (Years 1 10 )', 'FontSize',13, 'FontWeight','bold');
191 exportgraphics(fig4, fullfile(outputDir,'Fig04_Monthly_OPEX.png'),'Resolution',300);
192 fig5=figure('Position',[50 50 900 500],'Color','w');
193 cumNom = cumsum(cashOut);
194 cumDisc = cumsum(cashOut ./ dff);
195 plot(yy, cumNom/1e6, '-', 'Color', c1, 'LineWidth', 2); hold on;
196 plot(yy, cumDisc/1e6, '--', 'Color', c2, 'LineWidth', 2);
197 xlabel('Year');ylabel('M USD');title('Cumulative Pretreatment Cost');
198 legend('Nominal','Discounted','Location','nw');grid on;
199 exportgraphics(fig5,fullfile(outputDir,'Fig05_Cumulative.png'),'Resolution',300);
200 fig6=figure('Position',[50 50 1200 550],'Color','w');
201 co6=[c1;c2;c3;c4;c5;c6];
202 tl6= tiledlayout(1,2,'TileSpacing','compact','Padding','compact');
203 nexttile; hold on;
204 for eq = 1:nEQ
205     ydata = cumF(:,eq);
206     ydata(ydata <= 0) = 1;
207     plot(mx, ydata, 'Color', co6(eq,:), 'LineWidth', 1.8);
208 end
209 set(gca,'YScale','log');
210 ylim([1 max(cumF(:))]);
211 xlabel('Year');
212 ylabel('Cumulative Failures (log scale)');
213 title('Cumulative Equipment Failures');
214 legend(EQ(:,5),'Location','nw','FontSize',8);
215 grid on;
216 nexttile; hold on;
217 tp_=linspace(1,M,500);
218 for eq=1:nEQ
219     bt=EQ{eq,2}; et=EQ{eq,3}; Rs=exp(-(tp_/et).^bt);
220     plot(tp_/12, Rs*100, 'Color', co6(eq,:), 'LineWidth', 1.8);
221 end
222 xlabel('Year');ylabel('Survival (%)');title('Weibull Reliability Prediction');
223 legend(EQ(:,5),'Location','sw','FontSize',8);ylim([0 105]);
224 yline(50,'k--','50% life','LineWidth',1); grid on;
225 exportgraphics(fig6,fullfile(outputDir,'Fig06_Failures.png'),'Resolution',300);
226 fig7=figure('Position',[50 50 1100 500],'Color','w');
227 yyaxis left; plot(mx,mEn/1e3,'Color',[.7 .85 1],'LineWidth',.5); hold on;
228 ma12=movmean(mEn/1e3,12); plot(mx,ma12,'Color',c1,'LineWidth',2.5);
229 ylabel('Monthly Energy (MWh)');
230 yyaxis right;
231 plot(mx,mEP*100,'Color',c2,'LineWidth',2);
232 ylabel('Price (c/kWh)');
233 xlabel('Year');title('Dynamic Energy and Electricity Price');
234 legend('Monthly','12-mo avg','Price','Location','nw'); grid on;

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235 exportgraphics(fig7,fullfile(outputDir,'Fig07_Energy.png'),'Resolution',300);
236 fig8=figure('Position',[50 50 900 500],'Color','w');
237 plot(yx,aCI,'-o','Color',c5,'LineWidth',2,'MarkerSize',4); hold on;
238 p=polyfit(yx,aCI,1);
239 plot(yx,polyval(p_,yx),'--','Color',c2,'LineWidth',1.5);
240 xlabel('Year');
241 ylabel('kg CO_2/m^3');
242 title('Carbon Intensity Trend');
243 text(15,mean(aCI)+.02,sprintf('Slope: %.4f/yr',p_(1)),...
244     'FontSize',10,'Color',c2);
245 grid on;
246 exportgraphics(fig8,fullfile(outputDir,'Fig08_Carbon.png'),'Resolution',300);
247 fig9=figure('Position',[50 50 1100 500],'Color','w');
248 b10=bar(yx,[aE,aC,aFx,aF,aR]/1e6,'stacked','EdgeColor','none');
249 b10(1).FaceColor=c1;b10(2).FaceColor=c4;b10(3).FaceColor=c3;
250 b10(4).FaceColor=c2;b10(5).FaceColor=c5;
251 xlabel('Year');ylabel('M USD');title('Annual OPEX + Replacements');
252 legend('Energy','Chemicals','Fixed','Failures','Replacements','Location','nw');grid
    on;
253 exportgraphics(fig9,fullfile(outputDir,'Fig09_Annual_Cost.png'),'Resolution',300);
254 dataDir = fullfile(outputDir,'FigureData');
255 if ~exist(dataDir,'dir'), mkdir(dataDir); end
256 st = {'Raw','Screen','DAF','DMF','UF','Cart','RO_tgt'};
257 T = table(st, tp, sp, dp, 'VariableNames',{'Stage','Turb_NTU','TSS_mgL','SDI'});
258 writetable(T, fullfile(dataDir,'fig01_WQ.csv'));
259 T2 = table(mx, mTMP, mSDI, 'VariableNames',{'Year','TMP_bar','SDI'});
260 writetable(T2, fullfile(dataDir,'fig02_UF_TMP_SDI.csv'));
261 ix3 = 1:60;
262 T3 = table(mx(ix3), mT(ix3), mTurb(ix3), mTurbUF(ix3)*100, mCoag(ix3), ...
263     'VariableNames',{'Year','Temp_C','Turb_raw','Turb_UF_x100','Coag_mgL'});
264 writetable(T3, fullfile(dataDir,'fig03_Seasonal.csv'));
265 ix4 = 1:120;
266 T4 = table(mx(ix4), mEc(ix4)/1e3, mCc(ix4)/1e3, mFx(ix4)/1e3, mFc(ix4)/1e3, ...
267     'VariableNames',{'Year','Energy_k','Chemicals_k','Fixed_k','Failures_k'});
268 writetable(T4, fullfile(dataDir,'fig04_OPEX.csv'));
269 T5 = table(yy(:), cumNom(:), cumDisc(:), 'VariableNames',{'Year','CumNom','CumDisc'
    });
270 writetable(T5, fullfile(dataDir,'fig05_CumulativeCost.csv'));
271 EQnames_clean = {'Intake_pumps','DAF_saturators','DMF_vessels',...
272     'UF_modules','Chem_pumps','Cart_housings'};
273 T6_cum = array2table([mx, cumF], 'VariableNames',['Year', EQnames_clean]);
274 writetable(T6_cum, fullfile(dataDir,'fig06a_CumFailures.csv'));
275 tp_w = linspace(1,M,500)';
276 Rs_mat = zeros(length(tp_w), nEQ);
277 for eq=1:nEQ
278     bt=EQ{eq,2}; et=EQ{eq,3};
279     Rs_mat(:,eq) = exp(-(tp_w/et).^bt) * 100;
280 end
281 T6_wbl = array2table([tp_w/12, Rs_mat], 'VariableNames',['Year', EQnames_clean]);
282 writetable(T6_wbl, fullfile(dataDir,'fig06b_Weibull.csv'));
283 ma12 = movmean(mEn/1e3, 12);
284 T7 = table(mx, mEn/1e3, ma12, mEP*100, ...
285     'VariableNames',{'Year','Energy_MWh','Energy_MA12','Price_cents'});
286 writetable(T7, fullfile(dataDir,'fig07_Energy.csv'));
287 T8 = table(yx, aCI, 'VariableNames',{'Year','CarbonIntensity'});
288 writetable(T8, fullfile(dataDir,'fig08_Carbon.csv'));
289 T9 = table(yx, aE/1e6, aC/1e6, aFx/1e6, aF/1e6, aR/1e6, ...
290     'VariableNames',{'Year','Energy_M','Chemicals_M','Fixed_M','Failures_M','
    Replace_M'});
291 writetable(T9, fullfile(dataDir,'fig09_AnnualCost.csv'));
292 fprintf('Figure data exported to %s\n', dataDir);
293 fprintf('\n All figures saved to %s\n Done.\n',outputDir);

```

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